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Exploring local vegetables cultivated in home gardens of Pa Tio District, Yasothon Province, Thailand: ethnobotanical knowledge, nutritional value, and horticultural potential

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Abstract

Background Local vegetables are essential for food security, nutrition, and cultural practices in rural northeastern Thailand. Despite their importance, comprehensive documentation of their diversity, nutritional contributions, cultural significance, and economic value in Pa Tio District, Yasothon Province, is lacking. This study provides the first systematic account of edible plants in the district, highlighting their multifunctional roles in food, medicine, household nutrition, and local markets.

Methods Ethnobotanical data were collected through semi-structured interviews with local informants, and species uses were categorized as food, spices, medicinal, or ornamental. Quantitative indices—including the Cultural Importance Index (CI), Informant Consensus Factor (F_{ic}), Fidelity Level (FL), and Cultural Food Significance Index (CFSI)—were applied to assess species multifunctionality, frequency of use, and cultural prominence. Nutritional significance was evaluated using literature-reported mineral and vitamin contents, focusing on calcium, copper, iron, potassium, magnesium, manganese, sodium, phosphorus, zinc, and vitamin C. Economic assessments estimated annual income from marketable edible plants based on monthly prices, sales volume, and availability.

Results A total of 73 species from 33 families were recorded, with Cucurbitaceae, Zingiberaceae, and Apiaceae being particularly species-rich. Multifunctional species such as *Allium cepa* L., *A. sativum* L., and *Ocimum* spp. were central to diets, medicine, and local markets. Leafy greens and aromatic herbs were the most frequently used, contributing substantially to micronutrient intake. Marketable species provided supplementary income, enhancing household resilience and supporting local livelihoods.

Conclusion This study highlights the nutritional, socio-ecological and economic importance of edible plants in Pa Tio District. Conserving plant diversity and indigenous knowledge is essential for sustaining household nutrition and food security. Future work should investigate nutrient bioavailability, phytochemical composition, and improved

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cultivation practices to strengthen sustainable use and horticultural development of these culturally important edible plants.

Keywords Ethnobotany, Homegardens, Local vegetables, Pa Tio district, Yasothon province

Background

Local vegetables—encompassing traditional varieties, semi-domesticated species, and long-established cultivated crops—represent a core component of food systems, household livelihoods, and biocultural heritage throughout Southeast Asia. In rural communities, these plants provide essential micronutrients, dietary fiber, and diverse bioactive compounds that contribute significantly to nutrition and health. Beyond their dietary roles, local vegetables are tightly interwoven with culinary identity, traditional ecological knowledge, and community-based resource management [1]. Home gardens, in particular, serve as living repositories of these plant resources, functioning as multifunctional agroecosystems where food production, medicinal plant cultivation, and cultural practices converge [2]. Their ecological resilience, low input requirements, and continuous year-round productivity make them vital for household food security, especially in regions characterized by seasonal climatic variability [3].

Across neighboring countries in mainland Southeast Asia—such as Laos, Vietnam, Cambodia, and Myanmar—numerous studies highlight the indispensable role of home gardens in conserving local plant diversity and supporting subsistence diets. In Laos, for example, home-garden systems maintain a rich assemblage of leafy greens, climbers, herbs, and wild-derived species that contribute substantially to household nutrition and income generation [4]. Similar patterns are reported from Vietnam, where traditional vegetables persist as key sources of vitamins, antioxidants, and culturally significant ingredients despite increasing dietary modernization [5]. In Cambodia, ethnobotanical surveys show that villagers continue to rely on home-garden vegetables for both food and traditional medicine, reflecting strong cultural continuity and a high degree of local ecological knowledge [6]. Myanmar's rural communities likewise maintain diverse gardens that integrate food crops with medicinal and ritual plants, underscoring the multifunctionality of these systems across the region [7]. Collectively, these studies illustrate how home gardens serve as critical habitats for neglected and underutilized species (NUS), contributing to in situ conservation, resilience against food shortages, and the preservation of indigenous cultural practices [8].

Thailand shares many of these regional characteristics. Home gardens remain central to rural households, particularly in Northern and Northeastern (Isan) Thailand, where communities possess deep agricultural traditions

and extensive ethnobotanical knowledge related to edible plants [9, 10]. However, rapid socio-economic change, urban migration, market integration, and shifts toward commercial crop varieties have begun to alter local food systems. These transitions threaten the continuity of traditional edible plants use and may lead to the gradual erosion of plant genetic resources [11]. Although several studies in other parts of Thailand have examined home-garden diversity and ethnobotanical practices, research focusing specifically on the integrated nutritional, cultural, and horticultural value of edible plants in Yasothon Province remains limited [12]. Such knowledge gaps hinder the development of strategies that support both sustainable food systems and biodiversity conservation.

Pa Tio District in Yasothon Province provides a compelling context in which to examine these issues. The district is recognized for its strong reliance on home gardens, cultural persistence in using edible plants for both food and traditional medicine, and community engagement in small-scale agriculture. These characteristics make Pa Tio an important case study for understanding how edible plants contribute simultaneously to diet, health, cultural identity, and ecosystem sustainability. Yet, to date, a systematic assessment of the diversity, ethnobotanical uses, and nutritional significance of home-garden edible plants in this region has not been conducted. Comparative insights from Laos, Cambodia, Vietnam, and Myanmar further highlight the importance of documenting such knowledge before it is lost, as these countries face similar pressures associated with modernization, market-oriented agriculture, and environmental change [13].

The present study addresses existing knowledge gaps by investigating the diversity, ethnobotanical uses, and nutritional and economic significance of edible plants cultivated in home gardens of Pa Tio District, Yasothon Province. Specifically, the study aims to:

- (1) *Document the species diversity and composition of edible plants in home gardens across the district.*
- (2) *Examine their ethnobotanical roles, including their use as food, medicine, and other multifunctional purposes within the community.*
- (3) *Assess their contributions to household food security, taking into account nutritional value, cultural importance, and economic relevance.*
- (4) *Analyze patterns of species similarity among sub-districts to understand spatial distribution and diversity across the district.*

(5) Evaluate the economic value (EV) of commonly traded local vegetables based on market dynamics, availability, and annual income generated by vendors.

By integrating ecological, cultural, nutritional, and economic perspectives, this research provides a comprehensive understanding of home-garden systems and their importance for rural livelihoods. The findings aim to support the conservation of plant genetic resources, enhance local food systems, and inform sustainable strategies for promoting neglected and NUS and traditional vegetables in Thailand and the wider Southeast Asian region.

Methods

Study area

The study was conducted in Pa Tio District, located in Yasothon Province, northeastern Thailand, spanning approximate coordinates 15°47'N, 104°10'E (Fig. 1). The district is situated on the Khorat Plateau and experiences a tropical savanna climate, with three distinct seasons: a hot season from March to May, a rainy season from May to October, and a cool dry season from November to February. Average annual rainfall ranges from 1,300 to 1,600 mm, with most precipitation occurring during the monsoon period. The topography comprises lowland rice fields, upland areas, and scattered villages, supporting

diverse agroecosystems, including small-scale home gardens and cultivated fields [14].

Pa Tio District covers an area of approximately 308 km² and comprises five sub-districts—Chiang Pheng, Khok Na Ko, Krachai, Pho Sai, and Si Than. Soils are predominantly sandy loams with low fertility, characteristic of the Khorat Plateau, yet suitable for cultivating a wide range of edible plants. The district is primarily rural, with agriculture as the main livelihood. Rice farming dominates, often integrated with livestock raising and household gardening. Home gardens serve as multifunctional agroecosystems, providing vegetables, fruits, medicinal plants, and spices for daily consumption and local trade. These gardens are typically located adjacent to households and are mainly managed by women, who play a central role in maintaining ethnobotanical knowledge and seed exchange [16].

The population of Pa Tio District is approximately 18,500–19,000 inhabitants, predominantly composed of ethnic Lao-Isan communities. Lao-Isan is the main language, and Theravada Buddhism is the predominant religion. Local traditions, food culture, and rituals are closely tied to plant resources, reflecting the integration of cultural identity and traditional ecological knowledge. The combination of household gardening, small-scale agriculture, and access to nearby natural areas allows for dynamic interactions between local ecological

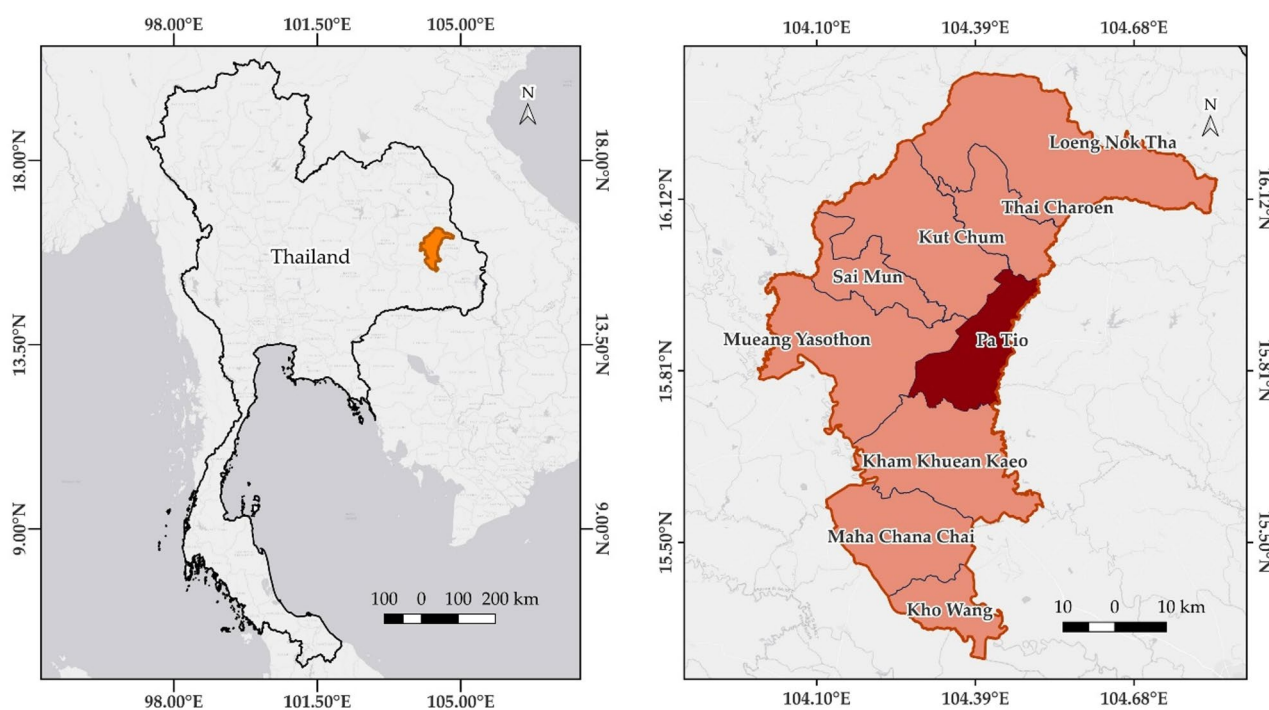


Fig. 1 Map of the study area: The left panel shows Yasothon Province highlighted in orange within Thailand, while the right panel emphasizes Pa Tio District in scarlet red within the province. (Map created with “QGIS” program ver. 3.34, geographic system ID: WGS 84; EPSG 4326 [15] designed by Phiphat Sonthongphithak)

knowledge, plant use, and socio-cultural practices. This setting provides a representative context to document and analyze the ethnobotanical knowledge of vegetables, spices, and other plant resources in rural northeastern Thailand [17].

Field survey and data collection

Field surveys were conducted between April 2024 and March 2025 in selected villages of Pa Tio District, Yasothon Province. A purposive sampling approach was employed to identify households with home gardens, focusing on those known to cultivate edible plants [18]. When selecting homegardens, we applied the following criteria: (1) proximity to the house; (2) clearly defined boundaries (e.g., fences of bamboo, bricks, or tree plantings); (3) a minimum area of 25 m²; and (4) active, continuous management without abandonment [19]. To ensure random sampling and minimize bias, we selected 14–21 well-preserved, traditionally managed homegardens in each subdistrict. In total, 80 households from five sub-districts were surveyed, selected purposefully based on the extensive presence of home gardens (Table 1).

Additionally, botanical identification was carried out through specimen collection and direct field observation. Scientific names were verified by Surapon Saensouk and cross-referenced using authoritative online resources, including Plants of the World Online (POWO) [20]. All confirmed specimens were deposited at the Vascular Plant Herbarium, Mahasarakham University (VMSU) in Kantharawichai District, Maha Sarakham Province, Thailand, for archival purposes and future study.

Ethnobotanical data collection

During the fieldwork, ethnobotanical data were collected through a two-step semi-structured interview process. In the first step, 30 home garden owners were interviewed to obtain general information on edible plants species cultivated in home gardens. The key questions included:

- (1) What kinds of plants do you have in your garden?
- (2) What are the main purposes or roles of these plants?
- (3) In what ways do you usually make use of these plants?

- (4) Which plant parts are most frequently employed for use?

In the second step, a broader set of 100 informants—comprising 50 women and 50 men, aged 20–75 years (Table 1), who were local residents born and raised in the area—were interviewed to capture detailed knowledge on plant use, cultural significance, and traditional practices related to edible plants. The informants included household members, traditional healers, and Buddhist monks, and were purposively selected to represent all five sub-districts of Pa Tio District, with 20 informants from each sub-district. This approach ensured comprehensive coverage of both ethnobotanical knowledge and household-level practices.

All interviews were conducted in Thai and the local Isan dialect, with the author—a native speaker of both languages—facilitating communication. No translator was required, ensuring accurate understanding of local terminology and nuanced ethnobotanical knowledge. The interviews focused on documenting plants cultivated in home gardens, including their local names, utilized plant parts, and preparation methods.

The interviews focused on documenting the ethnobotanical knowledge of plants cultivated in home gardens, including local names, plant parts used, and preparation methods. Prior to each interview, the objectives of the study were clearly explained, and verbal Prior Informed Consent (PIC) was obtained from all participants. Respondents were also informed of their rights, including voluntary participation and the ability to withdraw at any stage of the study.

All interviews were conducted in accordance with the ethical standards of the International Society of Ethnobiology (ISE) Code of Ethics [21] and followed the principles of the Nagoya Protocol on Access and Benefit Sharing [22].

Although the study did not involve sensitive personal data and formal institutional ethical approval was not required, all procedures were conducted in line with international standards for ethnobotanical research to ensure respect, transparency, and reciprocity toward participating communities. The interviews focused on documenting traditional knowledge, including local plant

Table 1 Number of households and demographic characteristics of surveyed sub-districts in Pa Tio District, Yasothon Province

Sub-district	GPS Coordinates		Number of households	Gen-der		Ethnicity	Language	Religion
	Latitude (N, S)	Longitude (E, W)		M	F			
Chiang Pheng	15°47'03"N	104°25'09"E	16	10	10	Lao Isan	Isan language	Christianity, Theravāda Buddhism
Khok Na Ko	15°55'17"N	104°26'01"E	17	10	10	Lao Isan	Isan language	Christianity, Theravāda Buddhism
Krachai	15°48'49"N	104°17'51"E	21	10	10	Lao Isan	Isan language	Christianity, Theravāda Buddhism
Pho Sai	15°50'30"N	104°23'48"E	12	10	10	Lao Isan	Isan language	Christianity, Theravāda Buddhism
Si Than	15°46'04"N	104°21'57"E	14	10	10	Lao Isan	Isan language	Christianity, Theravāda Buddhism

names, utilized plant parts, preparation techniques, and their respective uses.

Categorization of use-reports

Edible plants species documented in Pa Tio District were classified according to their reported uses into four categories: food (FD), medicinal (MD), ornamental (ON), and spice (SP). Food (FD) includes species consumed as staple vegetables or side dishes. Medicinal (MD) refers to plants utilized for therapeutic purposes or health maintenance. Ornamental (ON) covers species cultivated primarily for aesthetic, cultural, or ritual significance. Spice (SP) comprises plants used to enhance flavor or aroma in culinary preparations. During interviews, informants indicated all relevant uses for each species, and use-reports were recorded separately for each category. A single species could be assigned to multiple categories if it served more than one function, ensuring comprehensive representation of multifunctional plant use.

Nutritional analysis of edible plants

The nutritional composition of vegetables in Pa Tio District was assessed by reviewing published literature and established food composition databases. A total of 64 vegetable species commonly cultivated in homegardens were included in the analysis. For each species, data were compiled on mineral content (calcium, copper, iron, potassium, magnesium, manganese, sodium, phosphorus, and zinc) and vitamin C levels.

Nutritional information was retrieved from peer-reviewed journal articles, as well as from verified databases including PubMed, Web of Science, Scopus, and Google Scholar. Whenever nutritional data were available from studies conducted in Thailand, these values were prioritized. For species lacking local data, information from international studies was used. Data gaps or unavailable information were noted as missing (“-”) in the dataset.

The dataset summarizes the mineral and vitamin composition of each vegetable species, with nutrient values converted to mg/100 g when they were not originally reported in this unit.

Data analysis

Cultural importance index (CI)

The Cultural Importance Index (CI) assessment was conducted to evaluate the cultural significance of edible plants species used by the community in Pa Tio District, Yasothon Province. This approach applied quantitative ethnobotanical methods to capture the diversity of plant uses while ensuring accurate classification of utilization categories. Data were collected from local informants, with each plant species analyzed according to the number of reports across different use categories.

To calculate the CI, utilization reports were categorized by type of use, and the frequency of each report was recorded. The index was then determined using the following equation [23]:

$$CI = \sum_{u=u_1}^{u_{NC}} \sum_{i=i_1}^{i_{NC}} \frac{UR_{ui}}{N}$$

where CI denotes the cultural importance index of a plant species, UR represents the number of utilization reports for a given use category, I indicate the total number of use categories, NC refers to the total number of distinct utilizations, and N is the total number of informants. Thus, the CI reflects both the frequency and diversity of plant uses within the community, highlighting species that hold multifunctional cultural value and play integral roles in the daily lives and traditions of the Lao Isan people.

Plant part value (PPV)

The Plant Part Value (PPV) quantifies the frequency with which different plant parts are used in traditional practices, indicating which parts—such as leaves, fruits, roots, stems, or flowers—are most commonly utilized by the local community. Following Gomez-Beloz [24], PPV is calculated as:

$$PPV = \frac{\sum RU_{(plant\ parts)}}{\sum RU} \times 100$$

where $RU_{(plant\ parts)}$ represents the number of use reports for a specific plant part, and $\sum RU$ is the total number of use reports for all plant parts. PPV is expressed as a percentage, with higher values indicating more frequent use and greater significance of a plant part in traditional applications.

Informant consensus factor (F_{ic})

The Informant Consensus Factor (F_{ic}) assesses the degree of agreement among informants regarding the use of medicinal plants within a specific health category. It helps identify ailments for which the community holds a strong, shared knowledge of plant-based remedies. F_{ic} is calculated as [25]:

$$F_{ic} = \frac{N_{ur} - N_t}{N_{ur} - 1}$$

where N_{ur} is the total number of use reports for all species in a medicinal category, and N_t is the number of species cited within that category. F_{ic} values range from 0 to 1, with higher values indicating greater consensus among informants and a more focused, widely

recognized understanding of plant use for particular health conditions.

Fidelity level (FL)

The Fidelity Level (FL) quantifies the degree to which a plant species is specifically associated with a particular medicinal use among informants. It represents the proportion of people who report using a species for the same health condition, reflecting the specificity and consistency of its traditional application. FL is calculated as [26]:

$$FL = \frac{I_p}{I_u} \times 100$$

where I_p is the number of informants who cited the species for a specific ailment, and I_u is the total number of informants who mentioned the species for any medicinal purpose. Higher FL values indicate stronger agreement among informants and highlight a well-recognized medicinal role of the species within the community.

Cultural food significance index (CFSI)

The Cultural Food Significance Index (CFSI) was employed to evaluate the cultural importance of edible plant species used for both dietary and medicinal purposes. The index, first introduced by Pieroni (2001) [27], integrates seven dimensions of food significance into a single composite value, calculated as:

$$CFSI = QI \times AI \times FUI \times PUI \times MFFI \times TSAI \times FMRI \times 10^{-2}$$

The Quotation Index (QI) denotes the proportion of informants citing a particular species. The Availability Index (AI) reflects species abundance, from very common (4.0) to rare (1.0), while also considering the spatial scope of its use. The Frequency of Utilization Index (FUI) captures the regularity of consumption, ranging from more than once per week (5.0) to absence of use for over 30 years (0.5). The Parts Used Index (PUI) assesses the diversity of plant organs consumed, with values assigned to roots, stems, leaves, flowers, fruits, and seeds. The Multifunctional Food Use Index (MFFI) measures culinary versatility, encompassing preparation methods such as raw consumption, boiling, frying, roasting, and use as condiments. The Taste Score Appreciation Index (TSAI) reflects sensory evaluation, scored from 4.0 (very poor) to 10.0 (excellent). Finally, the Food-Medicinal Role Index (FMRI) evaluates the degree to which a plant is recognized as both food and medicine, ranging from “not recognized” (1.0) to “very high” (5.0).

To visualize patterns of cultural importance among edible plants, a heatmap was generated using PAST4 software (version 4.15). Each species was plotted against its

component CFSI indices (QI, AI, FUI, PUI, MFFI, TSAI, and FMRI) to show similarities and differences in cultural food significance. Hierarchical clustering was performed using the Unweighted Pair Group Method with Arithmetic Mean (UPGMA) on a similarity matrix derived from the normalized index values, highlighting groups of species with high, moderate, or low overall CFSI contributions. This approach provides a clear visual representation of clustering patterns and multifunctionality of vegetables within local diets [28].

Economic value (EV) assessment

The Economic Value (EV) assessment was conducted from a local economic perspective to analyze interactions between vegetable vendors and consumers in Pa Tio District, Yasothon Province. This approach integrated quantitative methods with botanical expertise to ensure precise identification of the scientific names of commonly traded vegetables.

A total of 80 homegardens were surveyed, of which 23 owners sold vegetables in local markets. Each vendor sold different vegetable species; therefore, the values used for EV calculation were averaged across vendors to obtain representative estimates. Market price data were collected monthly from April 2024 to March 2025.

To estimate the economic value, the mean price consumers were willing to pay for each species was calculated by averaging its maximum and minimum market prices. The annual economic value was then determined using the following equation:

$$EV = AP \times SM \times SP$$

where EV denotes the annual economic value of local vegetables, AP represents the average price per kilogram (THB/kg), SM indicates the total monthly sales volume (kg), and SP refers to the number of months the vegetables were available for sale during the year. Thus, the EV reflects the total yearly market value of local vegetables traded in the community, effectively representing the annual income vendors generated from selling each edible plant species [29].

Monthly surveys captured seasonal variations in availability, trade frequency, and price fluctuations. All data were recorded in pre-designed datasheets, and informant identities were anonymized to ensure ethical compliance and confidentiality.

Jaccard similarity index (JI)

To assess similarities in species composition among districts, the Jaccard Similarity Index (JI) was employed, a widely accepted measure for evaluating the degree of overlap between datasets. The index was calculated as follows [30]:

$$JI = a / (a + b + c)$$

where a represents the number of species shared by both districts, b the number of species unique to the first district, and c the number of species unique to the second.

Results

Diversity of edible plants in Pa Tio district

A total of 73 species of edible plants, representing 33 plant families, were documented in Pa Tio District (Figs. 2 and 3; Table 2). Among these, the family Cucurbitaceae exhibited the highest species richness with 8 species (10.96%), followed by Zingiberaceae with 7 species (9.59%) and Apiaceae with 6 species (8.22%). Both Fabaceae and Lamiaceae contributed 5 species each (6.85%), whereas Amaranthaceae and Solanaceae accounted for 4 species (5.48%) apiece. Amaryllidaceae and Piperaceae were represented by 3 species each (4.11%). Families represented by two species (2.74%) included Asteraceae, Convolvulaceae, Menispermaceae, and Rutaceae, while the remaining 20 families were represented by a single species (1.37%), reflecting the presence of less common but culturally recognized edible plants. The prominence of Cucurbitaceae, Zingiberaceae, and Apiaceae underscores their importance in household cultivation and routine consumption, whereas families with only one species illustrate the uneven yet broad distribution of edible plants diversity.

Of the 73 species, the majority were introduced species (48 species; 65.75%), while 25 species (34.25%) were native to the region (Table 2). This distribution suggests that introduced edible plants constitute a substantial portion of both local diets and cultivation practices, whereas

native species, although fewer, remain integral to traditional food systems.

An assessment of growth forms revealed that herbs were the most prevalent, accounting for 46 species (63.01%), followed by climbers (11 species; 15.07%), trees (9 species; 12.33%), and shrubs (4 species; 5.48%). Vines were the least represented, with only 3 species (4.11%). These patterns highlight the dominance of herbaceous species in home gardens, reflecting their ease of cultivation, short growth cycles, and frequent use in local diets.

Cultural importance index (CI) of edible plants in Pa Tio district

The Cultural Importance Index (CI) of edible plants recorded in Pa Tio District ranged from 0.11 to 1.12, reflecting substantial variation in cultural recognition and use among the species documented (Table 2).

The highest value was found in *Citrus × aurantiifolia* (CI=1.12), which is deeply embedded in local cuisine as a primary souring agent in soups, curries, and dipping sauces, while also being valued for its medicinal roles in digestive and respiratory ailments. Other key species with high CI values included *Allium cepa* (0.88), *Ocimum basilicum* var. *basilicum* (0.85), and *C. hystrix* (0.84). These species represent staple flavoring agents that are indispensable in household cooking. The inclusion of *Clitoria ternatea* (0.83) among the leading species highlights the importance of edible flowers, both as natural food colorants and for their perceived health benefits.

A second group of species, with CI values ranging from 0.68 to 0.82, comprised mainly aromatic herbs, spices, and versatile vegetables. This group such as *Ocimum tenuiflorum* (0.82), *Zingiber officinale* (0.82), *Allium sativum* (0.81), *Cymbopogon citratus* (0.81), *Carica*

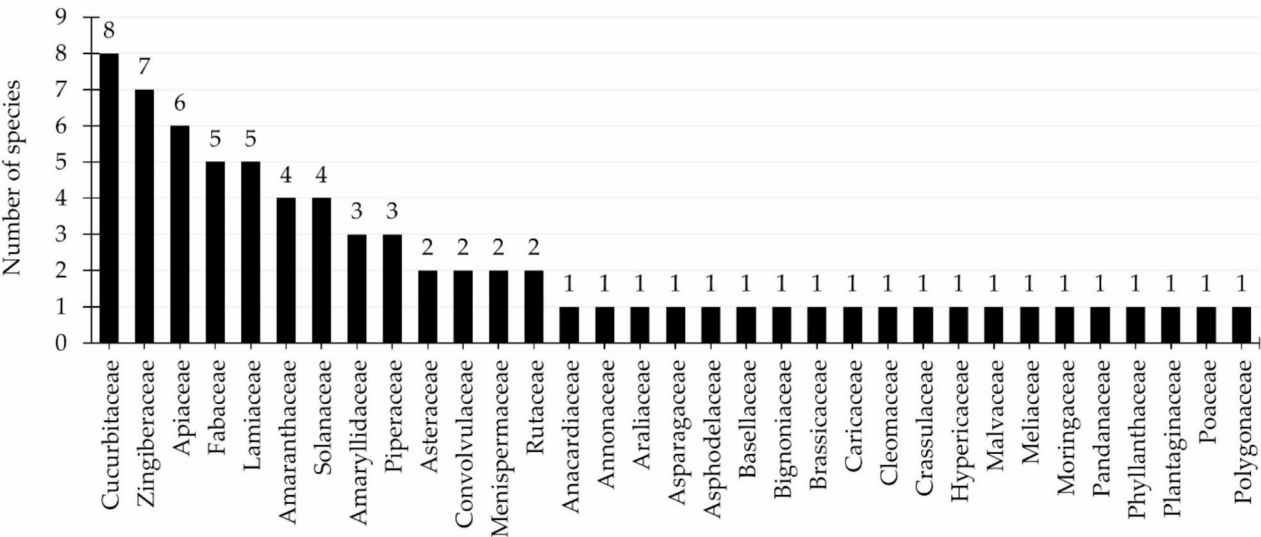


Fig. 2 Diversity of local edible plants documented in Pa Tio District, Yasothon Province



Fig. 3 Representative examples of edible plants documented in Pa Tio District, Yasothon Province. Photographs by Tammanoon Jitpromma

papaya (0.78), *Vigna unguiculata* subsp. *sesquipedalis* (0.78), *Capsicum annuum* (0.70), *Pandanus amaryllifolius* (0.68), and *Tiliacora triandra* (0.68). These taxa are widely recognized for their dual functions as staple food ingredients and as household remedies, particularly for conditions such as fever, digestive disorders, and

hypertension. The relatively high CI value of *Boesenbergia rotunda* (0.68) further underscores its cultural significance, reflecting its combined roles in both culinary practices and traditional medicine.

Species with moderate CI values (0.4–0.6), such as *Coriandrum sativum* (0.59), *Centella asiatica* (0.56),

Table 2 Ethnobotanical inventory of edible plants in Pa Tio District, including family, scientific name, vernacular name (Isan dialect), distribution in Thailand, growth habit, use-reports, cultural importance (CI), and collection number

Family	Scientific name	Vernacular name (Isan dialect)	Collection number	DT	GH	Use-reports				CI	UP	Mode of use
						FD	MD	ON	SP			
Amaranthaceae	<i>Amaranthus blitum</i> L.	Phak Khom Hat	TJY213	Introduced	Herb	35	-	-	-	0.35	LV, SH	Tender shoots are boiled or stir-fried with garlic; often added to clear soups or served as blanched vegetables.
Amaranthaceae	<i>Amaranthus viridis</i> L.	Phak Khom	TJY214	Introduced	Herb	37	-	-	-	0.37	LV, SH	Cooked similarly to <i>A. blitum</i> ; boiled in soups, fried with shrimp paste, or eaten blanched with chili dip.
Amaranthaceae	<i>Celosia argentea</i> L.	Ngonkai	TJY225	Introduced	Herb	-	-	11	-	0.11	WP	Cultivated in home gardens for its vibrant inflorescences; enhances aesthetic appeal of households; entire plant used for decorative purposes.
Amaranthaceae	<i>Gomphrena globosa</i> L.	Sam Pi	TJY244	Introduced	Herb	-	-	46	-	0.46	WP	Grown for bright, long-lasting flowers; used in home gardens and incorporated into "Bai Sri" arrangements for Buddhist rituals, demonstrating cultural and spiritual significance.
Amaryllidaceae	<i>Allium cepa</i> L.	Hom Daeng	TJY207	Introduced	Herb	22	30	-	36	0.88	BU, LV	Bulbs are sliced or chopped and incorporated into stir-fries, soups, curries, and sauces to enhance aroma and flavor; sometimes eaten raw with chili paste or added to salads for a pungent taste, while young leaves are occasionally used as garnish.
Amaryllidaceae	<i>Allium sativum</i> L.	Krathiam	TJY208	Introduced	Herb	12	34	-	35	0.81	BU, LV	Cloves are crushed, pounded, or finely chopped and fried as a base ingredient in curry pastes; stir-fried dishes, and soups to enhance aroma and flavor; sometimes eaten raw or mixed in dipping sauces, while fried garlic is also used as garnish.
Amaryllidaceae	<i>Allium tuberosum</i> Rottler ex Spreng.	Phak Paen	TJY209	Introduced	Herb	49	-	-	-	0.49	LV, IF	Leaves and flower stalks are chopped and stir-fried with eggs or meat; young inflorescences sometimes blanched and served with chili paste.
Anacardiaceae	<i>Spondias mombin</i> L.	Bak Kok	TJY271	Introduced	Tree	-	-	--	52	0.52	FT, LV	Young leaves eaten raw or in soups for tangy taste; fruits eaten fresh or added to curries to provide sourness; also made into preserves.
Annonaceae	<i>Annona squamosa</i> L.	Bak Khiab	TJY216	Introduced	Tree	53	-	-	-	0.53	FT	Ripe fruits eaten fresh or mixed with ice and sugar as a dessert; known for sweet, aromatic taste.
Apiaceae	<i>Anethum graveolens</i> L.	Phak Chilao	TJY215	Introduced	Herb	62	-	-	-	0.62	LV, SH, SD	Leaves are added at the end of cooking to flavor fish curries and soups; seeds occasionally used as spice.
Apiaceae	<i>Apium sellowianum</i> H. Wolff	Phak Khuen Chai	TJY217	Introduced	Herb	67	-	-	-	0.67	LV, SH	Leaves and stems used to flavor soups or stir-fries; also consumed fresh in salads.
Apiaceae	<i>Centella asiatica</i> (L.) Urb.	Buabok	TJY226	Native	Herb	18	38	-	-	0.56	LV, SH	Fresh leaves eaten raw with chili paste or blended into juice as a cooling tonic drink.
Apiaceae	<i>Coriandrum sativum</i> L.	Phakchi	TJY234	Introduced	Herb	-	-	-	59	0.59	LV, RT, SD	Leaves eaten raw as garnish in soups and curries; seeds roasted and ground into spice blends; roots pounded as essential ingredient in curry pastes.

Table 2 (continued)

Family	Scientific name	Vernacular name (Isan dialect)	Collection number	DT	GH	Use-reports				CI	UP	Mode of use
						FD	MD	ON	SP			
Apiaceae	<i>Cuminum cyminum</i> L.	Y'ia	TJY238	Introduced	Herb	-	-	-	34	0.34	SD	Seeds roasted and ground to enhance flavor in curry powders, stews, and fried dishes; used sparingly for warm, earthy aroma.
Apiaceae	<i>Eryngium foetidum</i> L.	Phak Hom Pe	TJY243	Introduced	Herb	54	-	-	-	0.54	LV	Aromatic leaves added to soups, curries, and spicy salads; eaten fresh for strong flavor.
Araliaceae	<i>Hydrocotyle umbellata</i> L.	Phak Waen Kao	TJY245	Introduced	Herb	31	-	-	-	0.31	LV, SH	Eaten fresh or lightly blanched with chili paste; used in salads for crisp texture.
Asparagaceae	<i>Dracaena fragrans</i> (L.) Ker Gawl.	Watsana	TJY242	Introduced	Shrub	-	-	61	-	0.61	WP	Valued for attractive foliage; commonly planted as household ornamental believed to bring prosperity and good fortune, especially when flowering.
Asphodelaceae	<i>Aloe vera</i> (L.) Burm.f.	Wan Hangchorakhe	TJY210	Introduced	Herb	-	35	-	-	0.35	LV, RT	Traditionally used to relieve burning sensations and abnormal vaginal discharge (leucorrhea).
Asteraceae	<i>Tagetes erecta</i> L.	Daorueang	TJY272	Introduced	Herb	-	-	21	-	0.21	WP	Planted for its decorative flowers; widely used in ceremonial arrangements and religious offerings, reflecting symbolic importance.
Asteraceae	<i>Zinnia elegans</i> Jacq.	Banchuen	TJY278	Introduced	Herb	-	-	17	-	0.17	WP	Grown for colorful blooms in home gardens; enhances visual appeal and contributes to ceremonial decoration.
Basellaceae	<i>Basella alba</i> L.	Phak Plang	TJY219	Native	Climber	43	-	-	-	0.43	LV, SH	Mucilaginous leaves boiled in soup or stir-fried with eggs; also eaten as blanched vegetable.
Bignoniaceae	<i>Oroxylum indicum</i> (L.) Kurz	Lin Fa	TJY258	Native	Tree	40	-	-	-	0.40	FT, IF	Young fruits and flowers boiled or cooked in curry; slightly bitter and aromatic.
Brassicaceae	<i>Brassica oleracea</i> L. "Phak Khana"	Phak Khana	TJY222	Introduced	Herb	65	-	-	-	0.65	LV, SH	Leaves boiled briefly or stir-fried with oyster sauce and garlic; commonly eaten with rice.
Caricaceae	<i>Carica papaya</i> L.	Bak Hung	TJY224	Introduced	Tree	78	-	-	-	0.78	FT	Unripe fruit shredded for <i>som tam</i> salad; ripe fruit eaten fresh or used in desserts.
Cleomaceae	<i>Cleome gynandra</i> L.	Phak Sian	TJY230	Native	Herb	55	-	-	-	0.55	LV, SH, IF	Leaves, shoot and inflorescence are pickled in brine or vinegar; also boiled in curries or stir-fried as sour-flavored vegetable.
Convolvulaceae	<i>Ipomoea alba</i> L.	Chom Chan	TJY246	Introduced	Herb	32	-	-	-	0.32	IF	Young flowers cooked in soups or stir-fried lightly; soft texture and mild taste.
Convolvulaceae	<i>Ipomoea aquatica</i> Forssk.	Phakbung	TJY247	Native	Herb	72	-	-	-	0.72	LV, SH	Commonly stir-fried with garlic or fermented shrimp; also boiled and served with chili dip.
Crassulaceae	<i>Kalanchoe blossfeldiana</i> Poelln.	Kulap Hin	TJY248	Introduced	Herb	-	-	18	-	0.18	WP	Appreciated for its bright, colorful flowers; cultivated in gardens for aesthetic enhancement.
Cucurbitaceae	<i>Benincasa hispida</i> (Thunb.) Cogn.	Fak	TJY220	Introduced	Climber	56	-	-	-	0.56	FT	Young fruit cooked in soups, stir-fries, or mild curries; flesh becomes soft and absorbs flavors.
Cucurbitaceae	<i>Coccinia grandis</i> (L.) Voigt	Phak Tamlueng	TJY232	Native	Climber	61	-	-	-	0.61	LV, SH	Young shoots and leaves boiled in curries or stir-fried; also eaten blanched with chili dip.
Cucurbitaceae	<i>Cucumis sativus</i> L.	Taeng Kwa	TJY236	Native	Vine	72	-	-	-	0.72	FT	Fruits eaten fresh with chili paste, in salads, or pickled; used to balance spicy dishes.

Table 2 (continued)

Family	Scientific name	Vernacular name (Isan dialect)	Collection number	DT	GH	Use-reports				CI	UP	Mode of use
						FD	MD	ON	SP			
Cucurbitaceae	<i>Cucurbita maxima</i> Duchesne	Fakthong	TJY237	Introduced	Climber	68	-	-	-	0.68	FT, IF, LV, SH	Young fruits, shoots, and flowers cooked in curries or stir-fries; ripe fruit boiled, mashed, or used in desserts.
Cucurbitaceae	<i>Luffa acutangula</i> (L.) Roxb.	Buap Liam	TJY250	Introduced	Climber	65	-	-	-	0.65	FT, LV	Tender fruits cooked in curries or soups; young leaves and shoots blanched and eaten with chili paste.
Cucurbitaceae	<i>Luffa aegyptiaca</i> Mill.	Buap Hom	TJY251	Introduced	Climber	62	-	-	-	0.62	FT, LV	Fruits boiled or stir-fried; shoots eaten as vegetable in curries or chili dips.
Cucurbitaceae	<i>Momordica charantia</i> L.	Mara Khi Nok	TJY253	Native	Climber	51	-	-	-	0.51	FT, LV	Bitter fruits sliced and stir-fried with eggs or cooked in curries; leaves boiled and eaten as side vegetable.
Cucurbitaceae	<i>Trichosanthes cucumerina</i> subsp. <i>cucumerina</i>	Buap Ngu	TJY274	Native	Climber	62	-	-	-	0.62	FT	Fruits sliced and cooked in curries or stir-fries; sometimes eaten raw in local salads.
Fabaceae	<i>Clitoria ternatea</i> L.	Anchan	TJY231	Introduced	Herb	35	43	5	-	0.83	IF, RT, WP	Flowers used to color rice, desserts, or beverages; also brewed as herbal tearmental purpose.
Fabaceae	<i>Psophocarpus tetragonolobus</i> (L.) DC.	Thua Phu	TJY265	Introduced	Climber	69	-	-	-	0.69	FT	Young pods eaten fresh, steamed, or stir-fried; used in salads and spicy dips.
Fabaceae	<i>Senegalia pennata</i> (L.) Maslin	Phak Kha	TJY266	Introduced	Tree	72	-	-	-	0.72	LV	Young shoots blanched or mixed in omelets; added to curries; distinct smell and slightly bitter.
Fabaceae	<i>Sesbania grandiflora</i> (L.) Poir.	Khae	TJY267	Introduced	Tree	64	-	-	-	0.64	LV, IF	Flowers boiled or stir-fried; used in soups and local curries; mild taste and soft texture.
Fabaceae	<i>Vigna unguiculata</i> subsp. <i>sesquipedalis</i> (L.) Verdc.	Thuaŋkyao	TJY275	Introduced	Vine	78	-	-	-	0.78	FT, SH	Long pods chopped and stir-fried with garlic or added to curries; shoots eaten boiled or fresh.
Hypericaceae	<i>Cratogeomys formosum</i> (Jack) Benth. & Hook.f. ex Dyer	Phak Tio	TJY235	Native	Tree	-	-	-	57	0.57	LV	Tender leaves boiled or blanched and added to soups or curries to impart a mild sour taste; sometimes eaten fresh with chili paste.
Lamiaceae	<i>Coleus amboinicus</i> Lour.	Hu Suea	TJY233	Introduced	Herb	47	-	-	-	0.47	LV	Aromatic leaves added to soups or minced into chili paste; used to mask strong meat odor.
Lamiaceae	<i>Mentha x villosa</i> Huds.	Saranae	TJY252	Introduced	Herb	75	-	-	-	0.75	LV	Leaves used as herbal tea or flavoring in drinks; added to salads for mild lemon aroma.
Lamiaceae	<i>Ocimum basilicum</i> var. <i>basilicum</i>	Horapha	TJY255	Native	Herb	85	-	-	-	0.85	LV	Leaves eaten raw or added to curries, stir-fries, and salads for aromatic flavor.
Lamiaceae	<i>Ocimum tenuiflorum</i> L.	Kra Phrao	TJY256	Native	Herb	-	-	-	82	0.82	LV	Fresh leaves added at the end of cooking in stir-fries, soups, and curries; also used in herbal teas for aroma and medicinal value.
Lamiaceae	<i>Ocimum x africanum</i> Lour.	I Tu	TJY257	Native	Herb	75	-	-	-	0.75	LV	Leaves used to season coconut curries and spicy dishes; also eaten fresh with fried foods.

Table 2 (continued)

Family	Scientific name	Vernacular name (Isan dialect)	Collection number	DT	GH	Use-reports				CI	UP	Mode of use
						FD	MD	ON	SP			
Malvaceae	<i>Abelmoschus esculentus</i> (L.) Moench	Krachap Khiao	TJY206	Introduced	Herb	58	-	-	-	0.58	FT	Young pods are sliced and cooked in curries or soups to create a slightly viscous texture; also stir-fried with chili and garlic or eaten blanched with chili paste.
Meliaceae	<i>Azadirachta indica</i> A.Juss.	Sadao	TJY218	Native	Tree	63	-	-	-	0.63	LV, IF	Bitter leaves and flowers are blanched and eaten with sweet chili sauce; believed to promote appetite and detoxify the body.
Menispermaceae	<i>Cissampelos pareira</i> L.	Khrua Ma Noi	TJY227	Native	Climber	49	-	-	-	0.49	FT	Leaf juice filtered into jelly form for cooking.
Menispermaceae	<i>Tiliacora triandra</i> (Colebr.) Diels	Yanang	TJY273	Native	Climber	-	-	-	68	0.68	LV	Leaves pounded and strained to extract mucilaginous juice used as natural thickener and flavoring agent in traditional curries such as <i>kaeng no mai</i> .
Moringaceae	<i>Moringa oleifera</i> Lam.	Phak I Hoom	TJY254	Introduced	Tree	43	-	-	-	0.43	FT, IF, LV	Young pods cooked in curries; leaves and flowers boiled in soups or fried; nutritious and widely used.
Pandanaceae	<i>Pandanus amaryllifolius</i> Roxb. ex Lindl.	BaiToei	TJY259	Introduced	Herb	-	47	-	21	0.68	LV	Fresh leaves tied into knots and added to rice and desserts for aroma; also used for wrapping and natural green coloring in sweets.
Phyllanthaceae	<i>Phyllanthus androgynus</i> (L.) Chakrab. & N.P.Balakr.	PhakWan Ban	TJY262	Native	Shrub	42	-	-	-	0.42	LV	Tender leaves boiled or stir-fried; often cooked in mild curries or soups; naturally sweet taste.
Piperaceae	<i>Peperomia pellucida</i> (L.) Kunth	Phak Ka Sang	TJY260	Introduced	Herb	39	-	-	-	0.39	WP	Whole plant used fresh in salads or blanched with chili paste; crisp texture and mild taste.
Piperaceae	<i>Piper nigrum</i> L.	Phrikthai	TJY263	Introduced	Vine	-	-	-	56	0.56	FT	Immature fruits boiled in soups and curries for spicy flavor; mature dried fruits ground into black or white pepper and used in sauces and marinades.
Piperaceae	<i>Piper sarmentosum</i> Roxb.	Phak I Loet	TJY264	Native	Herb	-	-	-	66	0.66	LV	Fresh leaves eaten raw with chili paste or used to wrap seasoned meat, fish, or noodles; sometimes shredded into salads for mild peppery flavor.
Plantaginaceae	<i>Limnophila aromatica</i> (Lam.) Merr.	Phak Kha Yaeng	TJY249	Introduced	Herb	32	-	-	-	0.32	LV, SH	Leaves and shoot used to flavor soups, curries, and spicy fish dishes; eaten raw with local salads.
Poaceae	<i>Cymbopogon citratus</i> (DC.) Stapf	Takhrai	TJY241	Introduced	Herb	-	36	-	45	0.81	LV, SH	Lower stems crushed or sliced and simmered in soups, curries, or teas for citrus fragrance; leaves used for herbal infusions and aromatic rice.
Polygonaceae	<i>Persicaria odorata</i> (Lour.) Sojak	Phak Phaeo	TJY261	Native	Herb	35	-	-	-	0.35	LV	Fresh leaves eaten with spicy soups or curries; also added to salads for distinctive aroma.
Rutaceae	<i>Citrus hystrix</i> DC.	Bak Kut	TJY228	Native	Shrub	-	42	-	42	0.84	FT, LV	Leaves torn and added to curries, soups, or chili paste for fragrance; rind grated or sliced finely for curry pastes and marinades to add citrus aroma.
Rutaceae	<i>Citrus × aurantiifolia</i> (Christm.) Swingle	Bak Nao	TJY229	Introduced	Shrub	-	41	-	71	1.12	FT, RT	Fresh fruits squeezed to add sour flavor in curries, soups, and dipping sauces; rind occasionally grated for pickles and pastes.

Table 2 (continued)

Family	Scientific name	Vernacular name (Isan dialect)	Collection number	DT	GH	Use-reports				CI	UP	Mode of use
						FD	MD	ON	SP			
Solanaceae	<i>Capsicum annuum</i> L.	Phrik	TJY223	Introduced	Herb	-	-	-	70	0.70	FT	Fruits are eaten fresh in spicy salads, dried and ground into chili powder, or fried in oil for sauces; intensity of use varies by household preference.
Solanaceae	<i>Solanum lycopersicum</i> L.	Bak Khuea Thet	TJY268	Introduced	Herb	52	-	-	-	0.52	FT	Fresh fruits eaten in salads or cooked in soups and curries to add sourness.
Solanaceae	<i>Solanum torvum</i> Sw.	Bak Khaeng	TJY269	Introduced	Herb	42	-	-	-	0.42	FT	Immature fruits cooked in curries or stir-fries; add texture and slight bitterness.
Solanaceae	<i>Solanum virginianum</i> L.	Makhuea Pro	TJY270	Introduced	Herb	67	-	-	-	0.67	FT	Fruits boiled or added to curries; used for flavoring and texture in traditional dishes.
Zingiberaceae	<i>Alpinia galanga</i> (L.) Willd.	Kha	TJY211	Native	Herb	-	-	-	51	0.51	RZ	Fresh or dried rhizomes are sliced or pounded and added to soups and curry pastes; also boiled with meat or fish to reduce odor and enhance aroma.
Zingiberaceae	<i>Alpinia siamensis</i> KSchum.	Kha	TJY212	Native	Herb	-	-	-	52	0.52	RZ	Rhizomes are sliced thinly and decocted for aromatic broth, or pounded and mixed in curry paste; locally believed to aid digestion and relieve gas.
Zingiberaceae	<i>Boesenbergia rotunda</i> (L.) Mansf.	Kra Chai	TJY221	Native	Herb	-	-	-	68	0.68	RZ	Rhizomes are crushed or thinly sliced and added to curry pastes, pickles, and chili sauces; also used in fried dishes for mild spiciness and aroma.
Zingiberaceae	<i>Curcuma angustifolia</i> Roxb.	Kra Chiao Daeng	TJY239	Native	Herb	46	-	-	-	0.46	IF	Inflorescences boiled or stir-fried as vegetable; mild aroma and soft texture.
Zingiberaceae	<i>Curcuma longa</i> L.	Khamin Chan	TJY240	Introduced	Herb	-	47	-	-	0.47	RZ	Traditionally used to relieve dyspepsia and digestive discomfort, including bloating, flatulence, nausea, and abdominal pain.
Zingiberaceae	<i>Zingiber montanum</i> (J. Koenig) Link ex A.Dietr.	Phlai	TJY276	Native	Herb	-	40	-	-	0.40	RZ	Traditionally used to purify the blood, regulate menstruation, and relieve digestive disorders such as bloating, abdominal pain, constipation, vomiting, and toothache.
Zingiberaceae	<i>Zingiber officinale</i> Roscoe	Khing	TJY277	Introduced	Herb	-	57	-	25	0.82	FT, RT, RZ	Fresh rhizomes sliced or crushed and added to stir-fries, soups, and curries; also boiled in tea or desserts for fragrance and health benefits.

DT, distribution in Thailand; GH, growth habit; FD, food; MD, medicine; ON, ornamental purposes; SP, spice; CI, cultural importance; UP, used parts; BU, bulb; FT, fruit; IF, inflorescence; LV, leaf; RT, root; RZ, rhizome; SD, seed; SH, shoot; WP, whole plant

Piper nigrum (0.56), and *Alpinia galanga* (0.51), play important but less universal roles. Their uses are often seasonal or dish-specific, suggesting that their cultural recognition is narrower compared to top-ranking vegetables.

At the lower end of the spectrum (<0.4), species such as *Amaranthus viridis* (0.37), *A. blitum* (0.35), *Aloe vera* (0.35), and *Persicaria odorata* (0.35) were recorded. These plants have specialized or less frequent uses, which may reflect changing dietary preferences or limited availability. Ornamental or rarely consumed species, including *Tagetes erecta* (0.21), *Kalanchoe blossfeldiana* (0.18), and *Zinnia elegans* (0.17), showed very low CI values, while *Celosia argentea* (0.11) represented the least culturally important vegetable in the study.

The analysis of Cultural Importance Index (CI) values across different categories of plant utilization in Pa Tio District revealed distinct patterns of cultural relevance. Food plants achieved the highest average CI value (0.53), underscoring their critical role in daily subsistence and nutrition. Closely following were spices (average CI=0.52), reflecting their dual importance as both culinary enhancers and culturally embedded household remedies.

Medicinal plants exhibited a lower average CI (0.41) compared to food and spice plants, yet still demonstrated substantial cultural significance, indicating their continued relevance in traditional healthcare systems. In contrast, ornamental plants recorded the lowest average CI (0.26), suggesting that while they are valued for aesthetic and cultural functions, they are less central to subsistence or healthcare practices.

Utilization of edible plants in Pa Tio district

The documented edible plants in Pa Tio District were classified into four primary utilization categories (Table 2; Fig. 4). The largest proportion belonged to the food category, comprising 47 species (55.29%), highlighting the central role of edible plants in household diets. The spice category ranked second with 19 species (22.35%), reflecting the cultural emphasis on flavor enhancement and condiment use in traditional cuisine. The medicine category included 12 species (14.12%), underscoring the importance of edible plants as sources of household remedies. The smallest category was ornamental, represented by 7 species (8.24%), indicating that certain edible plants are also valued for their aesthetic and cultural significance.

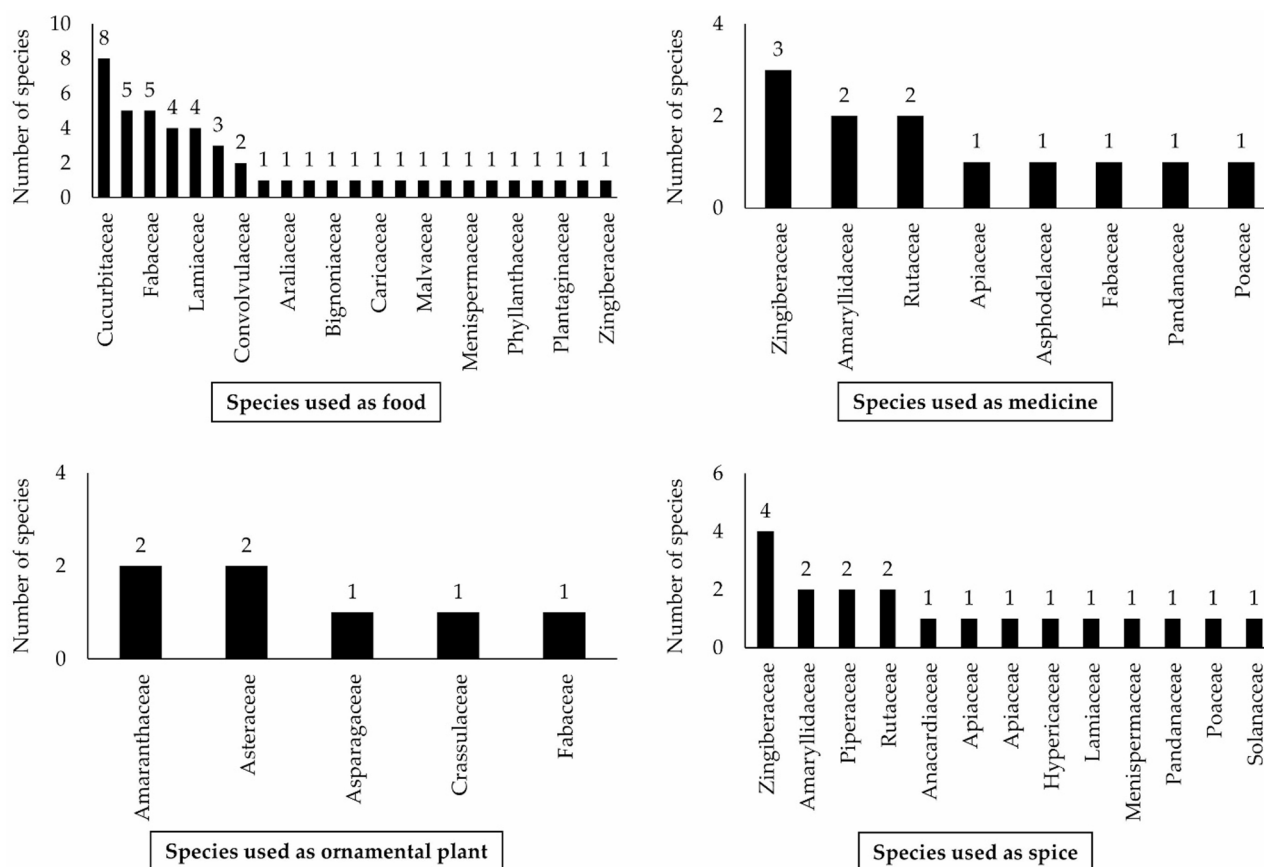


Fig. 4 Diversity of edible plant species and their utilization across four use categories in Pa Tio District, Yasothon Province

Edible plants used as food

A total of 47 species of edible plants, distributed across 23 families, were documented as being used for food in Pa Tio District (Table 2; Fig. 4). The Cucurbitaceae was the most dominant family, represented by 8 species (17.02%), reflecting its importance in local diets through commonly consumed gourds and squashes. Both the Amaryllidaceae and Fabaceae followed, with 5 species each (10.64%), highlighting their dual role as staple edible plants and flavoring agents in household cooking. Other well-represented families included the Apiaceae and Lamiaceae, each contributing 4 species (8.51%), primarily herbs and aromatic plants widely used for seasoning and culinary enhancement. The Solanaceae accounted for 3 species (6.38%), including commonly cultivated edible plants such as eggplants and tomatoes. The Convolvulaceae added 2 species (4.26%), notably *Ipomoea* species, which are staple leafy greens in the region. The remaining 16 families were represented by a single species each (2.13%), including Annonaceae, Araliaceae, Basellaceae, Bignoniaceae, Brassicaceae, Caricaceae, Cleomaceae, Malvaceae, Meliaceae, Menispermaceae, Moringaceae, Phyllanthaceae, Piperaceae, Plantaginaceae, Polygonaceae, and Zingiberaceae. While each contributes minimally in terms of numerical diversity, these families contain species of significant nutritional and cultural

importance, such as *Carica papaya*, *Centella asiatica*, and *Moringa oleifera*. Information on plant parts and preparation techniques is provided in Table S1.

The analysis of food plant utilization in Pa Tio District revealed that a wide range of plant parts were consumed (Fig. 5). Leaves were the most frequently used part (40.79%), highlighting their role as a staple ingredient in local diets. Fruits followed as the second most important part (22.37%), while shoots also contributed significantly (18.42%). Inflorescences accounted for 13.15% of uses, whereas bulbs (2.63%) and seeds (1.32%) were used less frequently. Additionally, one species (1.32%) was consumed as a whole plant.

Edible plants used as medicine

A total of 12 species of edible plants were recorded as being used for medicinal purposes in Pa Tio District (Table 2; Fig. 4). These species are distributed across 8 families, with the Zingiberaceae contributing the highest proportion (3 species; 25%). The Amaryllidaceae and Rutaceae followed, each represented by two species (16.67%). Families represented by a single species (8.33% each) included Apiaceae, Asphodelaceae, Fabaceae, Pandanaceae, and Poaceae.

The analysis of medicinal plant utilization in Pa Tio District showed that a variety of plant parts were used

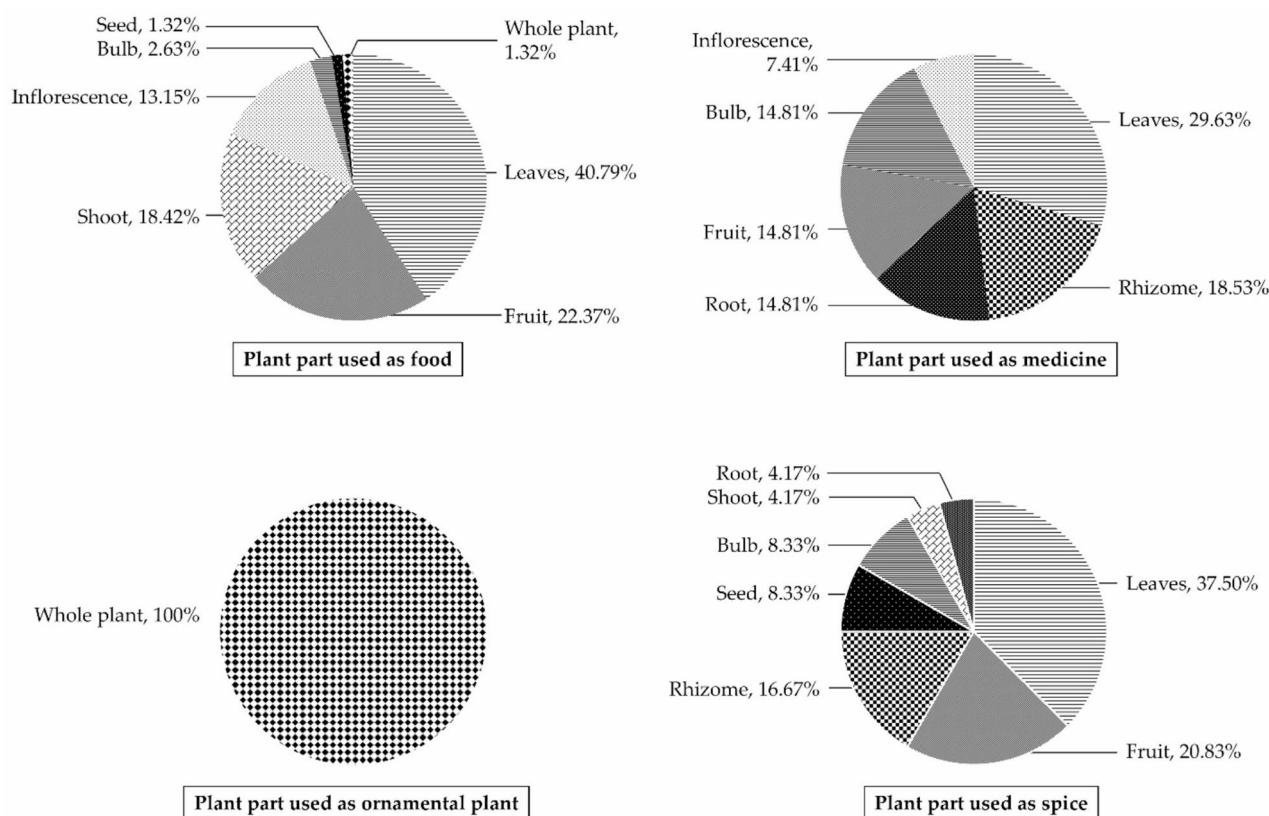


Fig. 5 Percentage of plant parts utilized across different categories of edible plants use in Pa Tio District, Yasothon Province

for therapeutic purposes (Fig. 5). Leaves were the most commonly used part (29.63%), emphasizing their importance in local healthcare practices. Rhizomes were the second most frequently employed part (18.53%), followed by bulbs, fruits, and roots, each accounting for 14.81% of uses. Inflorescences were less commonly utilized, representing 7.41% of citations.

Plants used for ornamental purposes

Flowering plants are frequently incorporated into home edible plants gardens to enhance their aesthetic value, particularly in households where gardens are situated at the front of the house. This practice not only improves the visual appeal of the household environment but also demonstrates the multifunctional roles of home gardens, which combine food production with cultural and decorative functions.

In Pa Tio District, a total of seven species of edible plants from five families were recorded as having ornamental value (Table 2; Fig. 4). The Amaranthaceae was the most represented family, with *Celosia argentea* and *Gomphrena globosa*, both cultivated for their vibrant inflorescences. The Asteraceae contributed to two species, *Tagetes erecta* and *Zinnia elegans*, which are widely grown for decorative purposes and play a role in ceremonial activities.

The remaining families were represented by one species each: Asparagaceae with *Dracaena fragrans*, valued for its attractive foliage and regarded as a symbol of good fortune; Crassulaceae with *Kalanchoe blossfeldiana*, appreciated for its colorful flowers; and Fabaceae with *Clitoria ternatea*, a multipurpose species notable for its edible, medicinal, and ornamental uses, particularly its striking blue blossoms.

Cultural significance was also noted. *Gomphrena globosa* and *Tagetes erecta* are often used in “Bai Sri” arrangements for Buddhist worship and other rituals, reflecting their symbolic importance. *Dracaena fragrans* is commonly planted as a household ornamental, with the belief that it brings prosperity and auspiciousness, especially when it flowers. These findings highlight that ornamental species in home gardens serve not only aesthetic functions but also hold deep cultural and spiritual significance in the local context. For ornamental purposes, the entire plant was utilized in all recorded species, representing 100% of the cases (Fig. 5).

Edible plants used as spice

In Pa Tio District, a total of 19 species of edible plants were identified as being used primarily as spices, belonging to 12 families (Table 2; Fig. 4). The Zingiberaceae family was the most represented, with 4 species (21.05%), underscoring its central role in flavoring and enhancing local cuisine. The Amaryllidaceae, Piperaceae, and

Rutaceae each contributed 2 species (10.53%), reflecting their widespread use in seasoning and traditional food preparations.

Other families were represented by a single species each (5.26%): Anacardiaceae, Apiaceae, Hypericaceae, Lamiaceae, Menispermaceae, Pandanaceae, Poaceae, and Solanaceae. These species are incorporated into dishes not only to improve taste but also to provide aroma, color, and in some cases, medicinal benefits, as many spices possess bioactive compounds with antimicrobial and digestive properties. Plant parts utilized and their preparation methods are summarized in Table S2.

The analysis of spice plants revealed that a variety of plant parts were utilized for seasoning purposes (Fig. 5). Leaves represented the most commonly used part (37.50%), reflecting their importance as a readily available and versatile source of flavor in local cuisine. Fruits were the second most frequently employed part (20.83%), followed by rhizomes (16.67%), which play a significant role as aromatic bases in traditional dishes. Bulbs and seeds were each cited at 8.33%, underscoring their specialized but important contributions to flavoring practices. Roots and shoots were less frequently used, each accounting for 4.17% of reported uses.

Fidelity level (%FL) of edible plants in Pa Tio district

The ethnomedicinal survey of edible plants in Pa Tio District revealed extensive traditional knowledge regarding the therapeutic applications of commonly cultivated species (Table S1). Informants reported diverse uses across eight primary therapeutic categories: Gastrointestinal, Infection, Parasite and Immune System, Musculoskeletal and Joint Diseases, Neurological system, Obstetrics, Gynaecology and Urinary Disorders, Poisoning and Toxicology, Respiratory Disorders, and Skin System. The parts most commonly used included bulbs, leaves, rhizomes, roots, fruits, and inflorescences, with preparation methods ranging from decoction and consumption to external application and extraction of juices.

Several species exhibited high cultural consensus for particular therapeutic uses, reflecting their perceived efficacy and entrenched role in local healthcare practices. *Allium cepa* demonstrated FL values of 46.67% for alleviating bloating and abdominal fullness and 53.33% for expelling intestinal parasites, indicating strong agreement among informants regarding its dual roles in gastrointestinal health and parasite management. Similarly, *A. sativum* showed FLs of 44.12% for stimulating appetite and enhancing digestive capacity, and 55.88% for anthelmintic effects, highlighting its multifunctionality and cultural importance in maintaining digestive health.

Leafy vegetables and herbs were frequently cited for musculoskeletal support and general vitality. *Centella asiatica* had FL values of 55.26% for treating internal

bruising and inflammation and 44.74% for alleviating fatigue and invigorating the body. *Pandanus amaryllifolius* exhibited FLs ranging from 29.79% to 36.17% across uses such as fatigue reduction, asthma management, and topical application as a facial mask, reflecting moderate consensus but demonstrating its multifunctional applications.

Several species were primarily associated with gastrointestinal health. *Curcuma longa* achieved FL values of 46.81% for relieving bloating, flatulence, abdominal pain, and digestive disturbances, and 53.19% for dyspepsia management. *Cymbopogon citratus* displayed FL values of 47.22% for fever reduction and 52.78% for supporting healthy digestion, highlighting both therapeutic consistency and cultural validation.

Plants used for infection control and immune system support also showed moderate to high FL values. *Zingiber officinale* exhibited FLs of 26.32% for supporting digestion, 33.33% for alleviating sore throat, and 40.35% for clearing respiratory passages, illustrating multifunctionality with moderate informant consensus. *Clitoria ternatea* had FLs of 20.93%, 37.21%, and 41.86% for its diuretic, antipyretic, and topical uses against insect stings, respectively, highlighting its versatility.

Reproductive and urinary health were supported by species such as *Zingiber montanum*, with FLs of 45% and 55% for menstrual regulation and gastrointestinal relief, respectively, and *Clitoria ternatea* as a traditional diuretic. Other species provided targeted treatments for neurological disorders (*Citrus hystrix* for dizziness and fainting) and skin conditions (*Aloe vera* for relieving burning sensation and topical application), further emphasizing the breadth of therapeutic applications in home gardens.

Overall, species with high FL values indicate strong cultural consensus and recognition of therapeutic efficacy, whereas moderate or lower FL values reflect either multifunctionality or divergent local practices. This highlights the complexity and richness of traditional

ethnomedicinal knowledge in Pa Tio District. Detailed information regarding plant species, parts used, preparation methods, therapeutic uses, and FL values is summarized in Table S1, providing a comprehensive framework for understanding the medicinal significance of edible plants.

Informant consensus factor (F_{ic}) of edible plants in Pa Tio district

The Informant Consensus Factor (F_{ic}) values were calculated to evaluate the level of agreement among informants regarding the medicinal uses of edible plants in Pa Tio District (Table 3). The F_{ic} values ranged from 0.961 to 1.000, indicating a generally high level of consensus across all therapeutic categories. The highest agreement ($F_{ic} = 1.000$) was observed for the neurological system, poisoning and toxicology, and respiratory disorders, each represented by a single species. High consensus values were also recorded for obstetrics, gynecology, and urinary disorders ($F_{ic} = 0.977$), as well as for the skin system ($F_{ic} = 0.970$). Categories with a broader range of species, such as infection, parasite, and immune system ($F_{ic} = 0.967$) and gastrointestinal disorders ($F_{ic} = 0.961$), also exhibited strong agreement among informants, reflecting the reliability and shared knowledge of these plants for treating common ailments. Similarly, the musculoskeletal and joint diseases category showed a high F_{ic} value of 0.961.

Cultural food significance index (CFSI) of edible plants in Pa Tio district

A total of 67 edible plant species were evaluated for their cultural food significance in Pa Tio District using the Cultural Food Significance Index (CFSI) (Table S2). The CFSI values varied widely among species, reflecting differences in frequency of utilization, availability, diversity of edible parts, multifunctional uses, taste appreciation, and medicinal relevance within the local food culture.

The species with the highest CFSI was *Allium cepa*, scoring 6150.00, highlighting its exceptional role as a staple ingredient and cultural food resource. This high value was attributed to strong performance across nearly all contributing indices, particularly frequency of utilization (FUI=5), taste score appreciation (TSAI=10), and food-medicinal relevance (FMRI=5). *Citrus × aurantiifolia* ranked second (CFSI=5925.00), followed by *A. sativum* (5625.00), *Ipomoea aquatica* (5082.00), and *Coriandrum sativum* (1696.00). These species demonstrated consistently high utilization frequency, excellent taste perception, and common integration into both dietary and folk medicinal practices.

Species with moderate CFSI values such as *Ocimum tenuiflorum* (912.00), *Citrus hystrix* (684.45), and *Coccinia grandis* (490.05), which are frequently consumed

Table 3 Informant consensus factor (F_{ic}) of local edible plants used as medicine in Pa Tio district

Therapeutic Categories	Number of Use Report (N_{ur})	Number of Taxa (N_t)	F_{ic}
Neurological system	20	1	1.000
Poisoning and Toxicology	18	1	1.000
Respiratory Disorders	16	1	1.000
Obstetrics, Gynaecology and Urinary Disorders	45	2	0.977
Skin System	34	2	0.970
Infection, Parasite and Immune System	151	6	0.967
Gastrointestinal	154	7	0.961
Musculoskeletal and Joint Diseases	52	3	0.961

in cooked dishes, condiments, or herbal teas. In contrast, species such as *Benincasa hispida* (67.50), *Peperomia pellucida* (38.61), and *Cuminum cyminum* (21.84) scored lower, suggesting that their roles are more limited, seasonal, or niche within local food systems.

The heat map analysis (Fig. 6) revealed clear clustering patterns of vegetable species based on their CFSI component indices. Species such as *Allium cepa*, *A. sativum*, and *Citrus × aurantiifolia*, clustered closely due to consistently high scores in availability, frequency of utilization, and medicinal relevance, confirming their strong integration into local foodways. Conversely, vegetables such as *Benincasa hispida*, *Luffa acutangula*, and *Trichosanthes cucumerina* subsp. *cucumerina* formed distinct low-value clusters, reflecting limited or specialized culinary roles.

Interestingly, several species with high PUI (Parts Used Index) and MFFI (Multifunctional Food Use Index), such as *Anethum graveolens* and *Clitoria ternatea*, demonstrated cultural versatility, being consumed in various forms ranging from fresh leaves to herbal infusions. This multifunctionality appears to contribute to their elevated CFSI values despite not ranking among the absolute top species.

Species with low FMRI (Food-Medicinal Role Index) generally had reduced overall CFSI scores, as they were less associated with traditional healthcare practices. However, plants such as *Centella asiatica* and *Pandanus amaryllifolius* balanced moderate scores across multiple indices, underscoring their dual role in diet and folk medicine.

The integration of ethnobotanical indices through the CFSI framework provides a holistic perspective on the cultural importance of vegetables in Pa Tio District. These findings highlight key species central to local diets and traditional knowledge, while also identifying underutilized plants with potential for future research, conservation, and sustainable food development initiatives in Northeastern Thailand.

Nutritional profile of local edible plants in Pa Tio district

The nutritional composition of 64 edible plant species was reviewed with a focus on mineral and vitamin content (Table 4). The dataset highlighted considerable variability in calcium (Ca), copper (Cu), iron (Fe), potassium (K), magnesium (Mg), manganese (Mn), sodium (Na), phosphorus (P), zinc (Zn), and vitamin C levels among species, reflecting their differential nutritional roles in local diets.

Analysis of the nutritional composition of edible plants revealed considerable variation in mineral and vitamin content. Calcium (Ca) was most abundant in *Citrus hystrix* (3987.45 mg/100 g) and lowest in *Moringa oleifera* (1.48 mg/100 g), while copper (Cu) peaked in *Solanum*

virginianum (598 mg/100 g) and was minimal in *Azadirachta indica* (0.002 mg/100 g). Iron (Fe) was highest in *S. virginianum* (1090 mg/100 g) and lowest in *Peperomia pellucida* (0.04 mg/100 g).

Potassium (K) reached maximum levels in *Basella alba* (7180.56 mg/100 g) and minimum in *Moringa oleifera* (1.75 mg/100 g). For magnesium (Mg), *Amaranthus blitum* (3042 mg/100 g) exhibited the highest concentration, whereas *Azadirachta indica* contained the least (0.038 mg/100 g). Manganese (Mn) was most concentrated in *Limnophila aromatica* (113.3 mg/100 g) and lowest in *Citrus × aurantiifolia* (0.01 mg/100 g).

Sodium (Na) was particularly high in *Solanum virginianum* (3750 mg/100 g) and low in *Sesbania grandiflora* (0.04 mg/100 g), whereas phosphorus (P) was maximal in *Moringa oleifera* (352.39 mg/100 g) and minimal in *Cucurbita maxima* (1.36 mg/100 g).

Zinc (Zn) showed the highest concentration in *Amaranthus blitum* (87.9 mg/100 g) and the lowest in *Peperomia pellucida* (0.08 mg/100 g). Finally, vitamin C was most abundant in *Basella alba* (1027.78 mg/100 g) and least in *Sesbania grandiflora* (0.99 mg/100 g).

These findings highlight those leafy vegetables, particularly in the Amaranthaceae and Lamiaceae families, are rich sources of multiple micronutrients, while certain climbers and herbs exhibit lower nutrient densities, emphasizing the importance of species diversity in local diets for optimal nutrition.

Economic value (EV) of local vegetables in Pa Tio district

During the survey of local vegetables in Pa Tio District, it was observed that, in addition to being cultivated for household consumption, several species are also sold in local markets (Table 5). This indicates that home gardens not only contribute to household nutrition but also provide a source of income for community members. To better understand the economic role of these vegetables, we assessed their market value, including price per kilogram, duration of availability throughout the year, and the average annual income generated by traders. Such an evaluation offers valuable insights into the contribution of local vegetables to rural livelihoods and highlights the importance of home gardens in supporting both food security and economic sustainability.

The economic analysis revealed considerable variation in price, market duration, and annual income for traders. Prices ranged from as low as 5 THB/kg for *Basella alba* to as high as 120 THB/kg for species such as *Allium sativum* and *Mentha × villosa*. Most edible plants were available for sale year-round, with 12 months of market availability, whereas a few, such as *Sesbania grandiflora*, were sold for only 6 months annually.

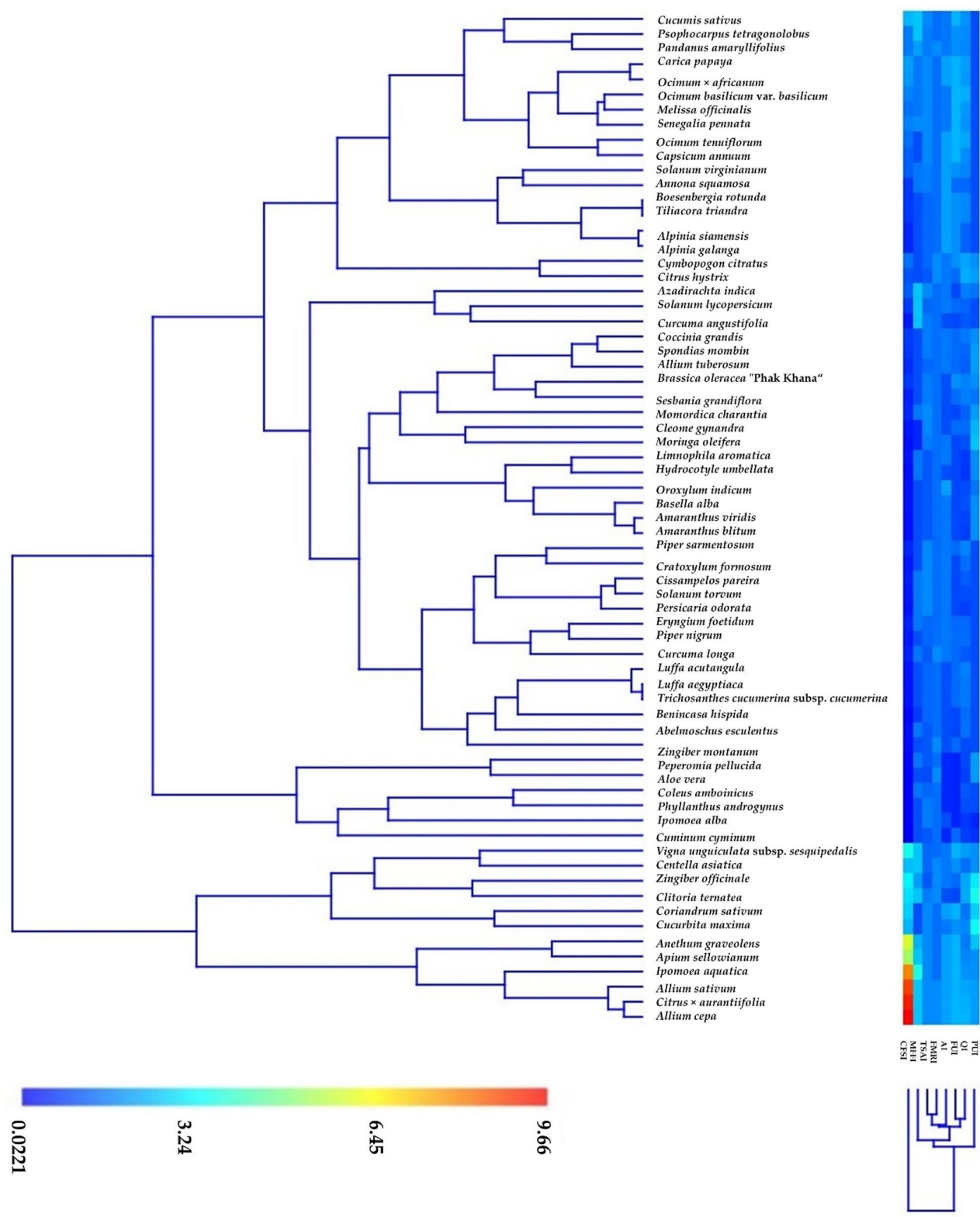


Fig. 6 Heat map analysis of local edible plants in Pa Tio District

Table 4 Nutritional profile of minerals and vitamins in some local vegetables (mg/100 g): a review of literature

No.	Scientific name	Ca	Cu	Fe	K	Mg	Mn	Na	P	Zn	Vit. C	References
1.	<i>Abelmoschus esculentus</i> (L.) Moench	265	0.49	1.6	340	102	0.14	110	-	2	-	[31]
2.	<i>Allium cepa</i> L.	151.47	-	2.91	2280.17	101.02	-	75.54	625.64	1.53	23.6	[32]
3.	<i>Allium sativum</i> L.	49.25	-	2.97	1451.31	63.61	-	57.85	605.8	3.42	4.49	[32]
4.	<i>Allium tuberosum</i> Rottler ex Spreng.	112.70	1.62	39.75	535.53	56.94	0.56	1.47	69.17	5.34	18.12	[33]
5.	<i>Aloe vera</i> (L.) Burm.f.	253.2	0.057	25	5818	831	11	2540	87.01	4.18	-	[34, 35]
6.	<i>Alpinia galanga</i> (L.) Willd.	139.43	-	3.97	2853.5	391.65	0.38	114.07	420.35	3.97	31.24	[32, 36]
7.	<i>Amaranthus blitum</i> L.	2474	2.61	115.4	1024	3042	-	20.75	-	87.9	184.815	[37]
8.	<i>Amaranthus viridis</i> L.	254.9	0.29	2.21	686	359	1.03	2.97	93	1.37	106.21	[38]
9.	<i>Anethum graveolens</i> L.	3170	-	41.6	1720	622	11.5	946	-	2.66	122	[39, 40]
10.	<i>Annona squamosa</i> L.	26.2	0.3	0.8	391	34	0.16	9	42	0.55	138.5	[40, 41]
11.	<i>Apium sellowianum</i> H.Wolff	23	2.4	6	1700	-	2.8	310	100	19	6.2	[31, 42, 43]
12.	<i>Azadirachta indica</i> A.Juss.	959.9	0.002	19.3	20	0.038	0.31	190	4.5	1.06	532.2	[44, 45]
13.	<i>Basella alba</i> L.	1097.22	0.56	58.33	7180.56	1027.78	-	347.22	-	2.78	1027.78	[46]
14.	<i>Benincasa hispida</i> (Thunb.) Cogn.	19	-	0.4	131	-	-	6	234.61	0.14	68	[47, 48]
15.	<i>Boesenbergia rotunda</i> (L.) Mansf.	141.99	-	6.96	2599.93	182.3	-	53.03	399.97	1.21	13.98	[32]
16.	<i>Brassica oleracea</i> L. "Phak Khana"	254	-	1.6	3362	169	2.7	400	55	6.1	254	[49, 50]
17.	<i>Capsicum annuum</i> L.	88.86	-	4.24	2679.49	158.24	-	84.64	438.33	1.81	8.29	[32]
18.	<i>Carica papaya</i> L.	24	-	0.1	257	10	-	3	5	-	48.12	[51, 52]
19.	<i>Centella asiatica</i> (L.) Urb.	176.7	0.02	4.1	377	104.4	12	21.2	32.2	10.9	107	[44, 53]
20.	<i>Cissampelos pareira</i> L.	69.6	-	0.96	12.6	-	-	-	-	-	-	[54]
21.	<i>Citrus hystrix</i> DC.	3987.45	-	9.18	1336.23	239.69	-	43.62	273.13	0.97	38.94	[32]
22.	<i>Citrus x aurantiifolia</i> (Christm.) Swingle	41.3	0.04	0.21	145	11.6	0.01	1.89	17.9	0.24	42.5	[55, 56]
23.	<i>Cleome gynandra</i> L.	594.8	0.1	21.7	1487.8	588.1	-	277.4	-	5.5	15.44	[57, 58]
24.	<i>Clitoria ternatea</i> L.	309.53	1.03	14.41	125.06	223.06	2.49	14.13	-	59.8	10.36	[59, 60]
25.	<i>Coccinia grandis</i> (L.) Voigt	457.81	0.44	39.69	4036.99	49.6	4.43	693.09	94	4.93	16.6	[61, 62]
26.	<i>Coleus amboinicus</i> Lour.	93	0.13	3.3	469.6	51.68	0.34	79.6	52.8	0.2	35.05	[63]
27.	<i>Coriandrum sativum</i> L.	1511.94	-	10.62	7050.99	204.8	-	457.11	585	2.94	118.3	[32]
28.	<i>Cratogeomys formosum</i> (Jack) Benth. & Hook.f. ex Dyer	448.3	-	17.2	-	-	-	-	-	-	395.4	[44]
29.	<i>Cucumis sativus</i> L.	139	2.49	7.8	437	70.8	0.79	151	-	5.27	2.8	[64, 65]
30.	<i>Cucurbita maxima</i> Duchesne	709	0.06	0.17	373.86	5.64	0.45	3.71	1.36	0.23	12.5	[36, 66]
31.	<i>Cuminum cyminum</i> L.	931	0.867	66.4	1790	366	3.33	168	499	4.8	7.7	[37]
32.	<i>Curcuma angustifolia</i> Roxb.	91.6	0.62	13.85	7.01	19.6	1.99	5.56	53.6	4.22	-	[67]
33.	<i>Curcuma longa</i> L.	23.24	0.0153	1.17	302.75	53.71	0.1446	0.706	46.97	-	1.12	[68, 69]
34.	<i>Gymnopogon citratus</i> (DC.) Stapf	178.74	-	2.1	2463.45	117.76	-	63.49	509.91	3.46	21	[32]
35.	<i>Eryngium foetidum</i> L.	1059.47	-	40.77	3679.61	207.74	-	240.28	672.72	4.37	36.36	[32]
36.	<i>Hydrocotyle umbellata</i> L.	-	0.02	-	-	-	-	-	-	0.3	-	[70]
37.	<i>Ipomoea aquatica</i> Forssk.	124.04	-	1.59	376.81	-	-	-	-	0.55	-	[71]
38.	<i>Limnophila aromatica</i> (Lam.) Merr.	992.68	7.8	25.77	1654.13	480.29	113.3	1965.19	407.45	-	38.7	[72]
39.	<i>Luffa acutangula</i> (L.) Roxb.	62	2.18	10.7	-	330	-	-	1050	5.8	205	[73]

Table 4 (continued)

No.	Scientific name	Ca	Cu	Fe	K	Mg	Mn	Na	P	Zn	Vit. C	References
40.	<i>Luffa aegyptiaca</i> Mill.	91.35	1.22	18.07	76.7	25.5	-	101.94	-	9.71	-	[74]
42.	<i>Mentha x villosa</i> Huds.	243	-	5.08	-	359.06	-	177.26	73	4.03	31.8	[75]
43.	<i>Momordica charantia</i> L.	188.1	-	13.8	-	-	-	-	-	-	460.6	[44]
44.	<i>Moringa oleifera</i> Lam.	148	0.45	25.14	1.75	301.11	7.21	133.11	352.39	2.04	-	[76]
45.	<i>Ocimum basilicum</i> var. <i>basilicum</i>	1898.32	-	9.11	3318.18	434.21	-	204.21	739.95	7.62	82.66	[32]
46.	<i>Ocimum tenuiflorum</i> L.	22	0.06	0.4	299	3.43	-	53.44	-	0.15	48.2	[77, 78]
47.	<i>Ocimum x africanum</i> Lour.	1774.88	-	9	2783.42	445.08	-	309.42	589.33	5.15	150.65	[32]
48.	<i>Oroxylum indicum</i> (L.) Kurz	3	0.41	0.3	235.3	14.2	0.2	0.5	31	6.4	-	[53]
49.	<i>Pandanus amaryllifolius</i> Roxb. ex Lindl.	-	2.65	8	41.1	94.82	1.85	-	-	1.39	-	[79]
50.	<i>Peperomia pellucida</i> (L.) Kunth	1.9	0.02	0.04	3.02	1.52	0.02	2.39	1.84	0.08	-	[80]
51.	<i>Phyllanthus androgynus</i> (L.) Chakrab. & N.P.Balakr.	70.5	-	0.9	-	-	-	396	128.2	0.9	56.1	[81]
52.	<i>Piper nigrum</i> L.	563.6	-	11.6	-	-	-	-	-	-	49.7	[44]
53.	<i>Piper sarmentosum</i> Roxb.	2218.58	1.58	41.53	5060.11	579.23	-	234.97	163.93	41.53	120.22	[46]
54.	<i>Psophocarpus tetragonolobus</i> (L.) DC.	730.84	-	6.97	2565.97	295.55	-	142.44	537.82	2.94	55.71	[32]
55.	<i>Senegalia pennata</i> (L.) Maslin	255.77	-	10.72	1903.47	233.53	-	80.71	1349.45	6.48	210.14	[32]
56.	<i>Sebania grandiflora</i> (L.) Poir.	80.2	-	13.5	381.45	-	-	0.04	290	-	0.99	[44, 73, 82]
57.	<i>Solanum lycopersicum</i> L.	163.33	-	12.5	93.33	73.33	0.06	164.5	178.33	0.57	20.52	[52, 83]
58.	<i>Solanum torvum</i> Sw.	358.57	-	3.38	2065.41	164.29	-	41.73	319.8	1.53	31.34	[32]
59.	<i>Solanum virginianum</i> L.	3550	598	1090	6750	110	2	3750	240	56	39.99	[84]
60.	<i>Spondias mombin</i> L.	11.04	0.12	0.33	288.28	15.1	0.03	5.55	-	-	-	[85]
61.	<i>Tiliacora triandra</i> (Colebrt.) Diels	155	0.25	7	309	-	-	11	50	0.42	31.26	[86, 87]
62.	<i>Trichosanthes cucumerina</i> subsp. <i>cucumerina</i>	2452.31	0.12	4.81	2472.77	1179.5	0.02	13.5	-	1.21	-	[88, 89]
63.	<i>Vigna unguiculata</i> subsp. <i>sesquipedalis</i> (L.) Verdc.	58.83	-	1.28	246.57	46.5	-	4.18	-	0.4	11.94	[52, 90]
64.	<i>Zingiber officinale</i> Roscoe	258.6	25.1	3.5	4709.64	264.62	-	169.33	592.77	2.74	20.83	[32, 91]

Ca, calcium; Cu, copper; Fe, iron; K, potassium; Mg, magnesium; Mn, manganese; Na, sodium; P, phosphorus; Zn, zinc; Vit. C, vitamin C; Data Unavailability is represented by "dash (-)"

Table 5 Economic value of local vegetables sold in Pa Tio district includes price range, market availability, and average annual income per trader

No.	Scientific name	Price THB/kg		Duration of Sales (Months per Year)	Average Annual Income (THB/a Trader)
		Max	Min		
1.	<i>Allium cepa</i> L.	80	78	12	20,856
2.	<i>Allium sativum</i> L.	120	110	12	34,500
3.	<i>Allium tuberosum</i> Rottler ex Spreng.	60	40	12	7,200
4.	<i>Alpinia siamensis</i> K.Schum.	60	40	12	11,400
5.	<i>Anethum graveolens</i> L.	60	40	12	9,000
6.	<i>Apium sellowianum</i> H.Wolff	70	30	12	13,800
7.	<i>Basella alba</i> L.	10	5	12	1,350
8.	<i>Benincasa hispida</i> (Thunb.) Cogn.	15	12	12	1,620
9.	<i>Boesenbergia rotunda</i> (L.) Mansf.	50	40	12	10,800
10.	<i>Brassica oleracea</i> L. "Phak Khana"	10	8	12	2,700
11.	<i>Capsicum annuum</i> L.	50	40	12	16,200
12.	<i>Carica papaya</i> L.	25	20	12	5,400
13.	<i>Citrus hystrix</i> DC.	50	45	12	5,700
14.	<i>Citrus × aurantiifolia</i> (Christm.) Swingle	30	25	12	3,300
15.	<i>Coriandrum sativum</i> L.	90	80	12	20,400
16.	<i>Cucumis sativus</i> L.	30	25	12	8,250
17.	<i>Cucurbita maxima</i> Duchesne	20	17	12	6,660
18.	<i>Cymbopogon citratus</i> (DC.) Stapf	15	10	12	4,350
19.	<i>Eryngium foetidum</i> L.	60	50	12	12,540
20.	<i>Luffa acutangula</i> (L.) Roxb.	24	18	12	5,292
21.	<i>Luffa aegyptiaca</i> Mill.	24	20	12	5,280
22.	<i>Mentha × villosa</i> Huds.	120	80	12	37,200
23.	<i>Ocimum basilicum</i> var. <i>basilicum</i>	40	25	12	11,700
24.	<i>Ocimum × africanum</i> Lour.	40	20	12	9,720
25.	<i>Persicaria odorata</i> (Lour.) Soják	50	40	12	14,040
26.	<i>Psophocarpus tetragonolobus</i> (L.) DC.	80	60	12	25,200
27.	<i>Senegalia pennata</i> (L.) Maslin	100	80	12	15,120
28.	<i>Sesbania grandiflora</i> (L.) Poir.	60	40	6	6,900
29.	<i>Solanum lycopersicum</i> L.	70	50	12	14,400
30.	<i>Trichosanthes cucumerina</i> subsp. <i>cucumerina</i>	50	40	12	13,500
31.	<i>Vigna unguiculata</i> subsp. <i>sesquipedalis</i> (L.) Verdc.	50	30	12	11,520

*Exchange rate — 1 USD = 32.37 THB (as of 25 November 2025)

Species generating the highest average annual income per trader included *Mentha × villosa* (37,200 THB), *Allium sativum* (34,500 THB), and *Psophocarpus tetragonolobus* (25,200 THB), reflecting both high market prices and consistent year-round availability. In contrast, species such as *Basella alba* and *Brassica oleracea* yielded lower annual incomes (1,350–2,700 THB) due to their low unit prices, despite being sold throughout the year.

Leafy herbs, spices, and specialty vegetables, including *Allium cepa*, *Ocimum* spp., and *Persicaria odorata*, provided moderate incomes ranging from 9,720 to 20,400 THB per trader annually, indicating steady demand in the local market. Cucurbitaceae species, such as *Cucumis sativus*, *Luffa acutangula*, and *Trichosanthes cucumerina*,

were also significant contributors to household income, with annual earnings between 5,280 and 13,500 THB.

Overall, these findings demonstrate that edible plants not only enhance household nutrition but also represent an important source of economic value for local farmers and traders. The combination of price, market duration, and annual income underscores the role of home-grown and locally marketed vegetables in sustaining rural livelihoods in Pa Tio District.

Discussion

Edible plants diversity

In Pa Tio District, a total of 73 edible plant species representing 33 families were documented, with Cucurbitaceae (10.96%), Zingiberaceae (9.59%), and Apiaceae (8.22%) being the most species-rich families. This pattern

is broadly consistent with ethnobotanical surveys in northeastern Thailand and Asia, where gourds, spices, and aromatic herbs are culturally and nutritionally prominent [92, 93]. Notably, 20 families were represented by a single species, reflecting specialized local knowledge and the retention of less common but nutritionally or medicinally significant taxa [94].

Compared with Saensouk et al. [95] in Mueang District, Yasothon Province, Pa Tio District shows similar patterns of family-level diversity, with a few dominant families and numerous single-species families, despite differences in overall species richness. This suggests a broadly comparable distribution of ethnobotanical knowledge and cultural significance across northeastern Thai communities.

Cultural importance of edible plants in Pa Tio district

The patterns of cultural importance observed in Pa Tio District highlight the centrality of multifunctional species in local diets and household healthcare practices. Edible plants that serve both culinary and medicinal roles—particularly aromatic herbs and spices—demonstrate how local ethnobotanical knowledge integrates nutrition, flavor, and therapeutic functions. These multifunctional species are consistently prioritized by the community, reflecting their embeddedness in daily practices and cultural traditions.

Comparable trends have been reported in other parts of Southeast Asia. For instance, Saensouk et al. [11] documented those edible plants such as *Ocimum* species and members of Zingiberaceae were highly valued in Mueang District, Yasothon Province for their dual roles in food and medicine, illustrating the cultural prominence of multifunctional taxa. Similarly, studies in rural Indonesia, Malaysia, Nepal, and Thailand have shown that edible plants used as flavoring agents or herbal remedies often hold greater cultural significance than staple edible plants consumed primarily for nutrition [96]. Together, these findings underscore that multifunctionality—combining taste, nutritional value, and health benefits—is a key driver of species prominence in traditional food systems.

Although ornamental species accounted for a smaller proportion of the edible plants assemblage, they were strategically incorporated into home gardens to enhance aesthetic and cultural value. This aligns with research on urban green spaces, where ornamental plants contribute not only to visual appeal but also to ecosystem services, mental well-being, and resilience to environmental changes [97]. In Pa Tio District, the inclusion of decorative species within kitchen gardens exemplifies a multifunctional approach in traditional agroecosystems, where plants simultaneously fulfill practical, nutritional, medicinal, and aesthetic roles.

Overall, these observations suggest that the drivers of cultural importance—multifunctionality, accessibility,

and integration into daily practices—are broadly consistent across Southeast Asian communities. The application of quantitative indices such as the Cultural Importance Index (CI) provides a robust framework to identify culturally essential species, offering valuable insights for conservation planning and strategies aimed at sustaining traditional knowledge systems [98]. By capturing both culinary and medicinal significance, this study contributes to a deeper understanding of the complex socio-ecological relationships that shape local food systems and community resilience.

Utilization of edible plants

The utilization patterns of edible plants in Pa Tio District align closely with broader ethnobotanical trends observed in northeastern Thailand and Southeast Asia. In this study, vegetables were categorized into four main use types: food (55.29%), spices (22.35%), medicine (14.12%), and ornamental purposes (8.24%). The predominance of food-related use mirrors findings from Saensouk et al. [95] in Mueang District, Yasothon Province, where more than half of the documented species were consumed as food. Similarly, the high proportion of spice species reflects the cultural emphasis on aromatic herbs and flavoring agents, consistent with studies from rural Indonesia and Malaysia, which identified Zingiberaceae and Amaryllidaceae as culturally important families for culinary enhancement [12].

Among food plants, Cucurbitaceae was the most represented family (17.02%), confirming their status as staple vegetables in northeastern Thai diets. Beyond their cultural importance, cucurbits are rich in macronutrients, minerals, and bioactive compounds with antidiabetic, antioxidant, and anticancer properties. Seeds, leaves, and flowers also provide additional health benefits, highlighting the multifunctionality of Cucurbitaceae species in supporting nutrition, health, and food security in Pa Tio District [99]. Amaryllidaceae and Fabaceae each contributed 10.64%, consistent with findings by Kebede et al. [100], who emphasized their dual roles in flavoring and nutrition. Apiaceae and Lamiaceae were moderately represented (8.51% each), reflecting the widespread use of herbs for seasoning, as previously reported in Tak provinces [101]. Leaves were the most frequently consumed plant part (40.79%), followed by fruits and shoots, which is similar to the utilization patterns reported by Thatantavide and Kumar [98], where leaves predominated in dietary use due to their accessibility and versatility.

In the medicinal category, Zingiberaceae was the most dominant family (25%), followed by Amaryllidaceae and Rutaceae (16.67% each). This confirms the recurrent importance of these families in ethnomedicinal systems throughout Southeast Asia [28]. High fidelity level (FL) values for *Allium cepa*, *A. sativum*, and *Curcuma longa*

parallel observations from previous ethnomedicinal studies in Maha Sarakham and Yasothon provinces, where these species were consistently cited for gastrointestinal and digestive ailments [11, 96]. Informant consensus factor (F_{ic}) values in the present study were uniformly high (0.961–1.000), reinforcing the reliability and widespread recognition of local medicinal knowledge, similar to patterns observed in neighboring country [102].

Spices accounted for 19 species across 12 families, with Zingiberaceae as the leading family (21.05%). This is consistent with prior studies indicating the ethnoculinary prominence of ginger, turmeric, and related species in northeastern Thai cuisine [28]. The diversity of plant parts used—leaves, fruits, rhizomes, bulbs, and seeds—also parallels earlier findings from Indonesia [103], highlighting the multifunctionality of spice species.

Although ornamental species represented a smaller proportion, their inclusion in home gardens demonstrates culturally embedded multifunctionality. Species such as *Dracaena fragrans*, *Gomphrena globosa*, and *Tagetes erecta* were cultivated not only for aesthetics but also for ritual and symbolic purposes, corroborating findings from studies in urban and rural agroecosystems in Thailand, where ornamental plants contribute to cultural expression, household identity, and ceremonial practices [93].

Cultural food significance and multifunctionality

Cultural Food Significance Index (CFSI) analysis highlighted key species central to both dietary and ethnomedicinal systems in Pa Tio District. *Allium cepa*, *A. sativum*, and *Citrus × aurantiifolia* achieved the highest CFSI values, reflecting their exceptional integration into daily consumption and traditional healthcare. These results are consistent with studies in Indonesia [104], where these species were also among the most frequently used and culturally valued edible plants in Yasothon Province.

Moderate CFSI species, including *Citrus hystrix*, *Coccinia grandis*, and *Ocimum tenuiflorum* reinforce previous reports highlighting the importance of aromatic herbs and indigenous greens as both culinary and medicinal resources [105]. Conversely, species with low CFSI values, such as *Benincasa hispida* and *Peperomia pellucida*, reflect either seasonal availability or niche uses, a pattern observed in ethnobotanical surveys from other Thai communities [106].

Species with high PUI (Parts Used Index) and MFFI (Multifunctional Food Use Index), such as *Anethum graveolens* and *Clitoria ternatea*, demonstrate cultural versatility and multifunctionality, being incorporated into diets as leaves, infusions, or flavoring agents. This multifunctionality parallels findings from studies in rural Malaysia and Indonesia, where species used for both

culinary and medicinal purposes often achieved higher cultural significance [96, 104].

Collectively, these patterns illustrate that multifunctionality, frequency of use, and perceived health benefits are major determinants of edible plants prominence in traditional food systems. Comparison with previous studies confirms that local ethnobotanical knowledge in Pa Tio District aligns with broader regional trends, highlighting a shared emphasis on culturally and nutritionally important species while also preserving specialized knowledge of less common taxa [107].

Nutritional contributions of edible plants

The nutritional diversity of edible plants in Pa Tio District underscores their pivotal role in supporting household dietary adequacy and mitigating micronutrient deficiencies [108]. Leafy greens, aromatic herbs, and fruiting edible plants together provide a complementary array of essential minerals and vitamins, emphasizing the necessity of dietary diversity over reliance on staple crops alone [109].

While certain species deliver functional benefits that support cardiovascular, muscular, and hematological health, others with lower nutrient densities highlight those familiar edible plants may not suffice to meet all micronutrient needs, reinforcing the importance of varied consumption [110]. This diversity also reflects the resilience of local food systems, fostering sustainable agricultural practices, adaptation to environmental variability, and preservation of traditional knowledge [111].

In this study, we focused on mineral and vitamin contents compiled from literature to illustrate the relative abundance of key nutrients across commonly cultivated kitchen garden edible plants. Although these data are derived from different sources and proximate compositions were not consistently available, they provide a useful overview of which species are rich in particular micronutrients, allowing the identification of nutritionally important edible plants for household consumption. This approach demonstrates nutritional complementarity within local diets rather than relying on a single dominant species [112].

The literature review also highlights important research gaps: several commonly cultivated edible plants remain understudied, and comprehensive analyses of essential minerals are still lacking for certain taxa, such as *Alpinia siamensis* and *Hydrocotyle umbellata*. Future research combining field-collected samples with standardized proximate and micronutrient analyses would strengthen these findings, validate ethnobotanical knowledge, and promote indigenous edible plants as sustainable, health-promoting food resources, thereby advancing both nutritional security and broader sustainable development goals.

Economic value market potential, and horticultural potential

The economic assessment of edible plants in Pa Tio District emphasizes their strategic role in supporting rural livelihoods beyond household nutrition. Home gardens function not only as sources of food but also as platforms for income generation, highlighting the intertwined nature of subsistence and market-oriented production in smallholder systems [8].

The diversity of marketable species reflects local dietary preferences and cultural values, with herbs, spices, and specialty edible plants demonstrating consistent consumer demand. These plants often occupy niche markets, where quality, freshness, and culinary or medicinal utility enhance their perceived value, creating stable economic opportunities for households engaged in their cultivation. Such dynamics suggest that local knowledge and culinary traditions directly influence economic outcomes in rural food systems [113].

Temporal availability of crops generally supports stable market income, as most species documented are present year-round. Only one species is not available throughout the year, as it is harvested for its flowers. It should be noted that data were collected from home garden owners, and due to the small size of these gardens, production may not always be sufficient for continuous sale throughout the year [114].

Beyond their economic importance, the findings highlight the horticultural potential of these edible plants. Many species demonstrate high adaptability to smallholder garden systems, ease of propagation, and resilience to local climatic conditions. Their multifunctional roles—including food, medicine, spice, and ornamental use—further enhance their suitability for home gardens and small-scale cultivation. This potential suggests opportunities for broader adoption in both household and community-level horticulture, contributing to agricultural diversification, food security, and sustainable rural development [115].

The findings further underscore the contribution of edible plants to diversified livelihood strategies. By integrating cultivation for both consumption and sale, households enhance resilience against economic shocks while simultaneously maintaining dietary diversity. This dual function positions edible plants as key instruments for promoting sustainable rural economies and reinforces the importance of supporting small-scale production systems through policy and extension services [116].

In summary, edible plants in Pa Tio District are not merely dietary resources; they represent an important supplementary economic and cultural asset. Although this study did not quantify household income, informants consistently emphasized that selling surplus edible plants provides small but steady financial support, helps cover

daily expenses, and reduces overall food costs. This qualitative evidence highlights the continued relevance of edible plants cultivation in sustaining household resilience and cultural practices.

Sociocultural and future trends influencing home-garden edible plants

Home gardens in Pa Tio District not only reflect traditional dietary and medicinal practices but are also shaped by broader sociocultural, economic, and environmental dynamics. Outmigration of younger community members for education or employment reduces available labor for garden maintenance, potentially influencing garden size and species composition. Concurrently, lifestyle changes and dietary shifts, including increased consumption of processed foods and commercially produced edible plants, may alter the frequency and diversity of edible plant use within households.

Agricultural intensification and market-oriented farming present both opportunities and challenges. While some households adopt improved cultivation techniques to increase yield and income, reliance on monocultures or hybrid varieties may limit the cultivation of traditional species. Conversely, growing consumer interest in local, culturally significant, and nutrient-rich edible plants may provide incentives to maintain and even expand home-garden diversity. Environmental factors, such as seasonal variability and climate change, can further affect planting schedules, flowering, and harvest timing, influencing both household consumption and potential market supply. Despite these pressures, the continued cultivation of multifunctional species—used simultaneously for food, medicine, ornamentation, and cultural purposes—demonstrates the resilience of traditional knowledge and the adaptive capacity of local communities.

Pa Tio District serves as a representative case study for other rural communities in northeastern Thailand. Even households with limited space often grow edible plants in small pots, integrating them as ornamental features while contributing to household nutrition. These practices, frequently supported by local leaders, encourage home-grown cultivation to reduce expenses and ensure access to safe, fresh produce. However, the prevalence, diversity, and impacts of such practices have not yet been systematically studied or documented.

Understanding these sociocultural, economic, and environmental trends in Pa Tio District is therefore essential not only for local planning but also for informing broader strategies in northeastern Thailand. Insights from this district can guide the conservation of culturally important species, support sustainable local food systems, and help preserve ethnobotanical heritage while anticipating future changes in home-garden composition.

Novel findings and implications

This study provides several novel insights into the ethnobotany of edible plants in Pa Tio District. First, the integration of nutritional, economic, and cultural data within a single framework highlights the multifunctionality of edible plants more comprehensively than previous surveys in northeastern Thailand [95, 98, 100]. Notably, species such as *Amaranthus blitum*, *Basella alba*, and *Ocimum × africanum* emerged as exceptional micronutrient sources while also supporting household income, underscoring their dual contribution to nutrition and livelihoods. Second, the combined use of the Cultural Food Significance Index (CFSI), Parts Used Index (PUI), and Multifunctional Food Use Index (MFFI) provided a more nuanced understanding of species prominence, highlighting that often underreported herbs and leafy vegetables—such as *Apium sellowianum*, *Cucumis sativus*, and *Ipomoea aquatica*—possess substantial cultural and nutritional value. Third, the documentation of ornamental plants integrated into home gardens for both aesthetic and cultural purposes add a new dimension to the study of multifunctional agroecosystems, reflecting the interplay between tradition, culture, and household management. Collectively, these findings expand current knowledge on the socio-ecological significance of edible plants and provide a robust evidence base for conservation and sustainable use.

Future research directions

Building on the findings of this study, several avenues for future research are recommended. First, while the mineral and vitamin composition of edible plants has been documented, studies on nutrient bioavailability and functional properties, including phytochemicals and antioxidant activity, are necessary to fully understand their health benefits. Second, longitudinal ethnobotanical surveys could capture seasonal and temporal variations in species availability, utilization patterns, and market dynamics, revealing shifts in cultural practices and household economies over time. Third, agroecological and conservation-focused research that examines the relationships between edible plants diversity, land-use patterns, soil fertility, and climate resilience would provide valuable guidance for sustainable cultivation and biodiversity protection. Fourth, socio-economic impact assessments quantifying the contribution of home-grown and market-traded edible plants to household income, food security, and resilience under socio-economic or environmental stressors would enhance understanding of their broader significance. Fifth, comparative studies across other districts in northeastern Thailand and neighboring countries could evaluate the generalizability of multifunctionality patterns and identify regionally important species for policy and conservation

prioritization. Finally, integrating edible plants use into public health and nutrition strategies could support culturally tailored interventions aimed at improving micronutrient intake and preventing non-communicable diseases. Addressing these research gaps will further strengthen the evidence base for promoting edible plants as a cornerstone of sustainable nutrition, livelihoods, and agro-biodiversity conservation.

Conclusions

This study documents a diverse assemblage of 73 plant species from 33 families in Pa Tio District, reflecting a rich ethnobotanical heritage and intricate biocultural relationships. The presence of both commonly cultivated and niche species underscores adaptive strategies that sustain household food security, nutrition, and cultural identity. Quantitative analyses indicate that multifunctional species, particularly *Allium cepa*, *A. sativum*, and *Ocimum* spp., occupy central roles in daily diets while simultaneously serving medicinal and cultural functions. High fidelity levels and cultural importance indices highlight strong community agreement on the therapeutic and culinary significance of key species, affirming the deep interconnection between traditional knowledge and local health practices. The variety of uses—from staple foods and spices to medicinal and ornamental purposes—demonstrates a holistic approach to agroecosystems, integrating multifunctionality with specialization. Such complexity supports dietary diversity, promotes nutritional adequacy, and enhances the resilience of rural food systems. Economically, the integration of home-grown and marketable species provides households with supplementary income, illustrating the dual role of edible plants in promoting both food security and livelihood sustainability. This reinforces the value of small-scale cultivation and the maintenance of traditional knowledge for socio-economic resilience. While this study did not include experimental validation, future research should focus on phytochemical analyses to identify bioactive compounds and on pharmacological studies to evaluate their biological activities. Such investigations would provide a scientific basis to support or critically assess the medicinal claims of traditional healers, thereby integrating indigenous knowledge with modern scientific approaches and promoting sustainable health solutions.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s13002-025-00841-w>.

Supplementary Material 1.

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Author contributions

Conceptualization, S.S., P.S. (Phiphat Sonthongphithak) and T.J.; methodology, P.S. (Piyaporn Saensouk), S.S., P.S. (Phiphat Sonthongphithak), K.C. and T.J.; software, P.S. (Phiphat Sonthongphithak) and T.J.; validation, P.S. (Piyaporn Saensouk), S.S., P.S. (Phiphat Sonthongphithak), K.C. and T.J.; formal analysis, T.J.; investigation, P.S. (Phiphat Sonthongphithak) and T.J.; resources, P.S. (Phiphat Sonthongphithak) and T.J.; data curation, T.J.; writing—original draft preparation, P.S. (Phiphat Sonthongphithak) and T.J.; writing—review and editing, P.S. (Piyaporn Saensouk), S.S., P.S. (Phiphat Sonthongphithak), K.C. and T.J.; visualization, T.J.; supervision, S.S.; project administration, T.J.; funding acquisition, S.S. All authors reviewed the final manuscript.

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Data availability

The original contributions presented in this study are contained within the article. Any further inquiries may be directed to the corresponding author.

Declarations

Ethics approval and consent to participate

Formal ethics approval was not required for this study under institutional guidelines at the time of research. All fieldwork, including market surveys and interviews, was conducted solely by the Thai authors, who are native speakers familiar with the local cultural context. Prior informed verbal consent was obtained from all participants after clearly explaining the study's objectives, methods, and voluntary nature in the local language. No personal or sensitive information was collected, and participants were informed of their right to withdraw at any time. All procedures complied with the ethical principles of the International Society of Ethnobiology (ISE) Code of Ethics and adhered to the Nagoya Protocol on Access and Benefit Sharing.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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